MIL-M-28787/64A(EC)
1 May 1980
SUPERSEDING
MIL-M-28787/64(EC)
22 June 1976

MILITARY SPECIFICATION

* MODULES, STANDARD ELECTRONIC

GATE, AND-OR-INVERT, DIGITAL, KEY CODE LHJ

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the procurement of an electronic digital AND-OR-INVERT gate module.

1.2 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table II.

Name		Gate, AND-OR-INVERT
Family		Digital
Key code		LHJ
Size		1A
Weight		35 grams maximum
Failure rate		0.30 failures/10^6 hours maximum
* Environment		Class I as specified in MIL-STD-1389
Maximum power dissipation		660 mW (outputs not loaded)
Supply voltage		5 Vdc ±10 percent
Input voltage		
Low level		-0.5 V minimum, 0.8 V maximum
High level		2.0 V minimum, V_{CC} maximum
Input current		
Low level	(V_{IL} = 0.0 V)	-2.2 mA maximum
	(V_{IL} = 0.5 V)	-2.0 mA maximum
High level	(V_{IH} = 2.5 V)	50 µA maximum
	(V_{IH} = 5.5 V)	1.0 mA maximum
Output voltage		
Low level	(I_{OL} = 20 mA)	0.5 V maximum
High level	(I_{OH} = 1.0 mA)	2.5 V minimum

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command, ATTN: ELEX 5043, Department of the Navy, Washington, D.C. 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

* U.S. GOVERNMENT PRINTING OFFICE: 1980—603 121/2547

FSC 5963

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:49Z
Check the source to verify that this is the current version before use.

Output drive current maximum
capability
E1 thru E6
Low level (V_{OL} = 0.5 V) 18 mA
High level (V_{OH} = 2.5 V) -0.85 mA

F1 thru F6
Low level (V_{OL} = 0.5 V) 20 mA
High level (V_{OH} = 2.5 V) -1.0 mA

Propagation delay time
t_{PLH} inputs to E outputs
C_L = 50 pF 5.5 ns minimum, 15 ns maximum
C_L = 150 pF 7.0 ns minimum, 18 ns maximum

* t_{PLH} inputs to F outputs
C_L = 50 pF 5.5 ns minimum, 21 ns maximum
C_L = 150 pF 7.0 ns minimum, 29 ns maximum

t_{PHL} inputs to E outputs
C_L = 50 pF 3.5 ns minimum, 10.5 ns maximum
C_L = 150 pF 5.5 ns minimum, 15 ns maximum

t_{PHL} inputs to F outputs
C_L = 50 pF 5.5 ns minimum, 16.5 ns maximum
C_L = 150 pF 7.5 ns minimum, 21 ns maximum

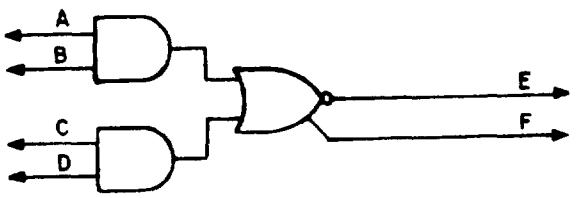
* 1.3 Module function. This module contains six independent AND-OR-INVERT functions, with an additional inverter on each output as illustrated on figure 1.

1.3.1 Input to output relationship. Each circuit has the input to output relationship of table I for positive logic.

TABLE I. Input to output relationship.

Inputs				Outputs	
A	B	C	D	E	F
L	L	L	L	H	L
L	L	L	H	H	L
L	L	H	L	H	L
L	L	H	H	L	H
L	H	L	L	H	L
L	H	L	H	H	L
L	H	H	L	H	L
L	H	H	H	L	H
H	L	L	L	H	L
H	L	L	H	H	L
H	L	H	L	H	L
H	L	H	H	L	H
H	H	L	L	L	H
H	H	L	H	L	H
H	H	H	L	L	H
H	H	H	H	L	H

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:49Z
Check the source to verify that this is the current version before use.



$E = \overline{AB + CD}$
$F = AB + CD$

NOTES:
1 H = High-voltage level.
2 L = Low-voltage level.

1.3.2 Parallel outputs. Paralleling of outputs is not permissible except when both inputs and outputs of circuits are paralled to obtain extended drive. Each circuit whose output is paralleled with an output of another circuit must have each of its inputs paralleled with a corresponding input of the other circuit. Circuits may be paralleled only with other circuits within the same group. These circuit groups are:

a. Circuits 1 and 2.
b. Circuits 3 and 4
c. Circuits 5 and 6.

The output drive capability of this combination is equal to the sum of the output drive capabilities of the individual circuits. The input loading of this combination is equal to the sum of the input loading of the individual circuits.

1.3.3 Crosstalk relationship.

* 1.3.3.1 Module crosstalk. When connected in the worst case mode of operation for the F outputs (all but one F output switching simultaneously from a high level to a low level) the LHJ module could exhibit substantial crosstalk. When the quiet F output line is at a low level there may be as much as a 1,000 mV positive going spike coincident with the negative edge of the switching F outputs.

1.3.3.2 Circuit crosstalk. The crosstalk on any F output pin is nearly directly proportional to the number of simultaneously switching F outputs. For every switching F output in group A (circuits 1, 3, and 4) the noise induced on a low level quiet F output in group A is as much as 190 mV. For every switching F output outside group A (circuits 2, 5, and 6) the noise induced on a low level quiet F output in group A is as much as 70 mV.

For every switching F output in group B (circuits 2, 5, and 6) the noise induced on a low level quiet F output in group B is as much as 175 mV. For every switching F output outside group B (circuits 1, 3, and 4) the noise induced on a low level quiet F output in group B is as much as 90 mV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:49Z
Check the source to verify that this is the current version before use.

* 2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARD

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

* 3.3 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 0.30 failures per million operating hours when predicted in accordance with MIL-STD-1389.

* 3.4 Performance requirements. The module shall conform to the electrical requirements specified in table II and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be LHJ.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:49Z
Check the source to verify that this is the currentversion before use.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5. Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/64-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance test. All production modules shall be tested in accordance with notes 1 and 2 of table II.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed two years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table II (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table II for a power supply voltage (V_{CC}) of 5.0 ±0.5 Vdc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 Vdc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

* 4.3 Test equipment. The following test equipment is necessary to conduct acceptance tests for this module. Test equipment shall be as specified in MIL-STD-1665 and that listed below:

Equipment	MIL-STD-1665 designator	No. required (maximum per test figure)
DC power supply	Item P01	4
Pulse generator	Item J01	1
Megohm bridge	Item R06	1
DC voltmeter	Item V03	1
DC ammeter	Item X01	1
Pulse generator	Item J05	1
Oscilloscope	Item D05	1
Oscilloscope probe	Item Y01	2
Capacitance meter	Item C05	1
Capacitance bridge	Item C01	1

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:49Z
Check the source to verify that this is the current version before use.

Resistors	No. required (maximum per test figure)
249Ω ±1%, 1/4 W	6
277Ω ±1%, 1/4 W	6
1 kΩ ±5%, 1/2 W	1
51Ω ±5%, 1/2 W	2
590Ω ±1%, 1/4 W	1
Value as required for figure 12, note 2, F outputs, 1/4 W	6
Value as required for figure 12, note 2, E outputs, 1/4 W	6
2.49 kΩ ±1%, 1/8 W	6
2.94 kΩ ±1%, 1/8 W	6
Capacitors	
Value as required for figure 12, note 4, test type S	12
Value as required for figure 12, note 4, test type Q	12
Diode	
1N3600, 1N4150, or equivalent	12

All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table II.

* 4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity (SEMP-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through X shall be performed in any order with the following exceptions. If conditions A, B, C, D, E, or N are part of a sequence of tests, the order of testing shall be as follows: A, N, B, C, D, E, where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A thru X in table II shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through W (except N) shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table II which are marked with an "X" in the test type column, and test condition W, shall be performed using the initial limits at 25° ±5°C ambient.

Source: https://assist.dla.mil -- Downloaded: 202640-20T15:49Z
Check the source to verify that this is the current version before use.

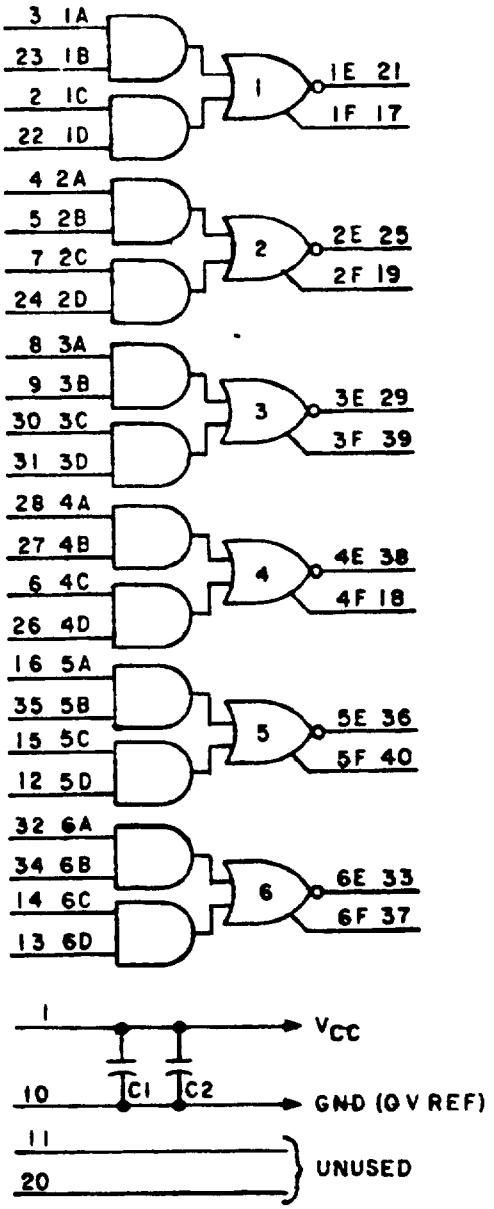


FIGURE 1. LHJ module function diagram.

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:49Z
Check the source to verify that this is the current version before use.

TABLE II. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25° ± 5°C Ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation	III	S	10	∞	1.0	∞	10	∞	1.0	∞	MΩ
B1-B12	V_{OUT} (low-loaded)	IV	S	0.01	1.0	0.01	1.0	0.01	1.0	0.01	1.0	V
C1-C24	Input breakdown current	V	S	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	mA
D1-D12	Output short circuit current	VI	X	-40	-100	-40	-100	-40	-100	-40	-100	mA
E1-E6	Output leakage current	VII	X	0.0	4.0	0.0	4.0	0.0	4.0	0.0	4.0	mA
E7-E12	Output leakage current	VII	X	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	mA
F1-F24	Input clamp diode voltage	VIII	X	-0.4	-1.2	-0.4	-1.2	-0.4	-1.2	-0.4	-1.2	V
G1-G24	Static output voltage, high level	IX	X	2.5	4.5	2.5	4.5	2.5	4.5	2.5	4.5	V
H1-H12	Static output voltage, low level	IX	X	0.0	0.5	0.0	0.5	0.0	0.5	0.0	0.5	V
J1-J16	Functional	X	X	Test vectors must produce output vectors as specified in table X								
K1-K24	Input current, low level	XI	S	-0.1	-2.2	-0.1	-2.2	-0.1	-2.2	-0.1	-2.2	mA
L1-L24	Input current, high level	XII	X	0	50	0	50	0	50	0	50	µA
M	Power supply current, low level outputs	XIII	X	40	120	40	120	40	120	40	120	mA
N1, N2	Unused pin isolation	XIV	S	10.0	∞	1.0	∞	10.0	∞	1.0	∞	MΩ
P1-P11 (odd)	Transition time (t_{TLH}), C_L = 50 pF	XV	S	2.0	17.0	2.0	17.0	2.0	20.0	2.0	20.0	ns
P2-P12 (even)	Transition time (t_{TLH}), C_L = 50 pF	XV	S	2.0	21.0	2.0	21.0	2.0	24.0	2.0	24.0	ns

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:49Z
Check the source to verify that this is the current version before use.

TABLE II. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25° ± 5°C Ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
P13-P23 (odd)	Transition time (t_{TLH}), C_L = 150 pF	XV	Q	2.0	32.0	2.0	32.0	2.0	33.0	2.0	33.0	ns
P14-P24 (even)	Transition time (t_{TLH}), C_L = 150 pF	XV	Q	2.0	32.0	2.0	32.0	2.0	35.0	2.0	35.0	ns
Q1-Q12	Transition time (t_{THL}), C_L = 50 pF	XV	S	2.0	9.0	2.0	9.0	2.0	10.0	2.0	10.0	ns
Q13-Q24	Transition time (t_{THL}), C_L = 150 pF	XV	Q	2.0	19.0	2.0	19.0	2.0	20.0	2.0	20.0	ns
R1-R24	Propagation delay time (t_{PLH}), inputs to E outputs, C_L = 50 pF	XVI	S	3.5	15.0	3.5	15.0	3.5	17.0	3.5	17.0	ns
* R25-R48	Propagation delay time (t_{PLH}), inputs to F outputs, C_L = 50 pF	XVI	S	5.5	21.0	5.5	21.0	5.5	23.0	5.5	23.0	ns
R49-R72	Propagation delay time (t_{PLH}), inputs to E out puts, C_L = 150 pF	XVI	Q	5.0	18.0	5.0	18.0	5.0	20.0	5.0	20.0	ns
R73-R96	Propagation delay time (t_{PLH}), inputs to F outputs, C_L = 150 pF	XVI	Q	7.0	27.0	7.0	27.0	7.0	31.0	7.0	31.0	ns
S1-S24	Propagation delay time (t_{PHL}), inputs to E outputs, C_L = 50 pF	XVI	S	3.5	10.5	3.5	10.5	3.5	12.5	3.5	12.5	ns

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 20260-20T15:49Z
Check the source to verify that this is the current version before use.

TABLE II. Module electrical requirements - Continued.

	Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25° ± 5°C Ambient temperature				Operating temperature 0° to 60°C				Unit
					Initial		End-of-life		Initial		End-of-life		
					Min	Max	Min	Max	Min	Max	Min	Max	
	S25-S48	Propagation delay time (t_{PHL}), inputs to F outputs, C_L = 50 pF	XVI	S	5.5	17.0	5.5	17.0	5.5	19.0	5.5	19.0	ns
	S49-S72	Propagation delay time (t_{PHL}), inputs to E outputs, C_L = 150 pF	XVI	Q	4.5	18.0	4.5	18.0	4.5	20.0	4.5	20.0	ns
	S73-S96	Propagation delay time (t_{PHL}), inputs to F outputs, C_L = 150 pF	XVI	Q	6.5	24.5	6.5	24.5	6.5	26.5	6.5	26.5	ns
*	T1-T6	Output crosstalk, low level	XVII	Q	0	750	0	750	0	750	0	750	mV
*	T7-T12	Output crosstalk, low level	XVII	Q	0	1000	0	1000	0	1000	0	1000	mV
*	T13-T18	Output crosstalk, low level	XVII	Q	0	500	0	500	0	500	0	500	mV
	U1-U12	Output crosstalk, high level	XVII	Q	0	400	0	400	0	400	0	400	mV
	V	Power supply transient amplitude	XVIII	Q	0	500	0	500	0	500	0	500	mV
	W	Power supply filter capacitance	XIX	S	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	µF
	X1-X24	Input capacitance	XX	Q	2.0	15.0	2.0	15.0	2.0	15.0	2.0	15.0	pF

1/ "X" Parameters shall be 100 percent tested during module acceptance tests.
2/ "S" Parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" Parameters shall be tested during qualification testing only.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:49Z
Check the source to verify that this is the current version before use.

* 4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through W shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table II. Following the return of the module to 25°C, electrical tests D through X (except N) shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table II.

5. PACKAGING

* 5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table II, the module is considered to have completed its useful life (see 3.8).

6.3 Cross reference of classification. Part number M28787/64-1 supersedes part number M28787/64. The two parts are interchangeable.

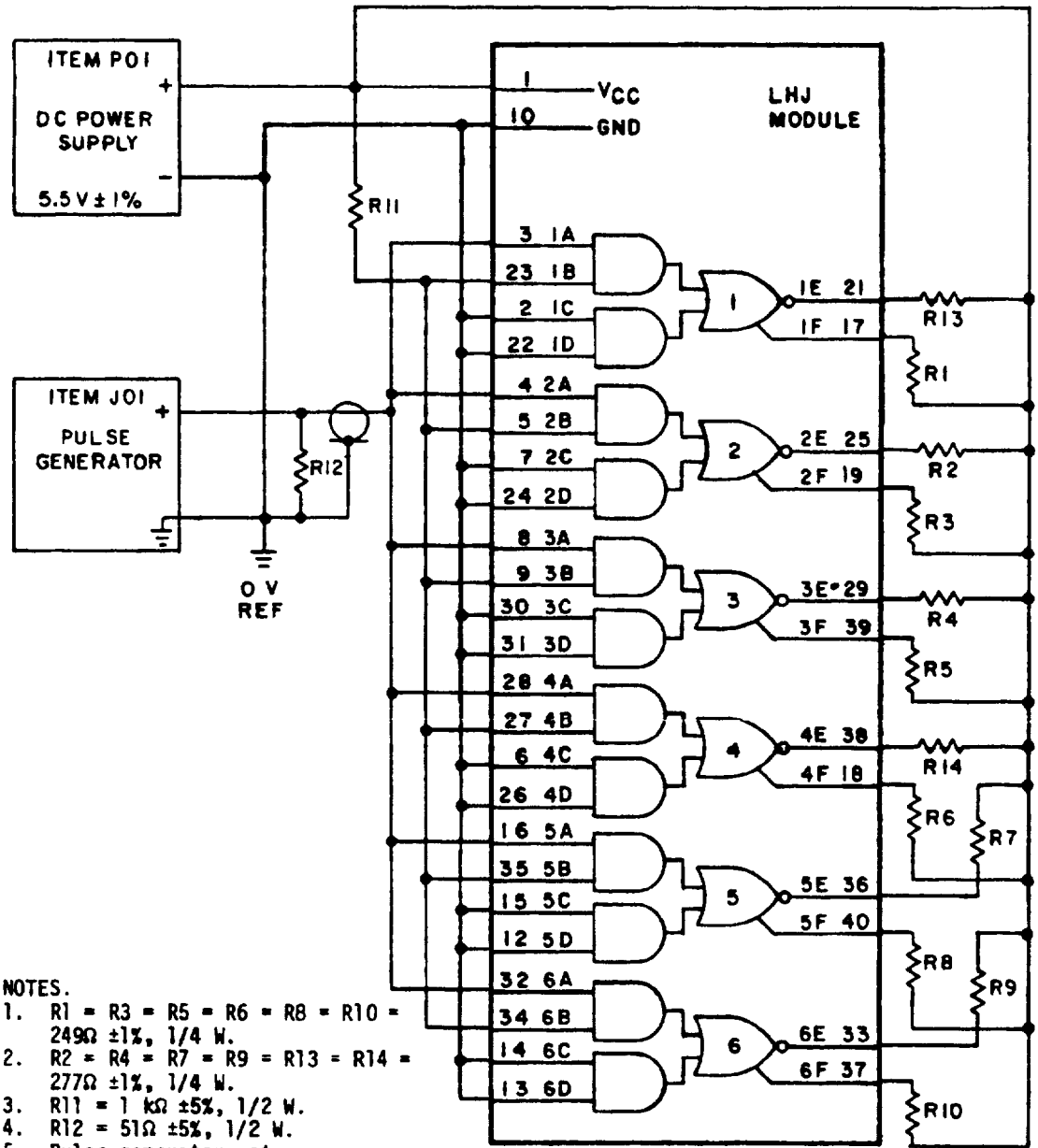
6.4 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5963-N114)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:49Z
Check the source to verify that this is the current version before use.

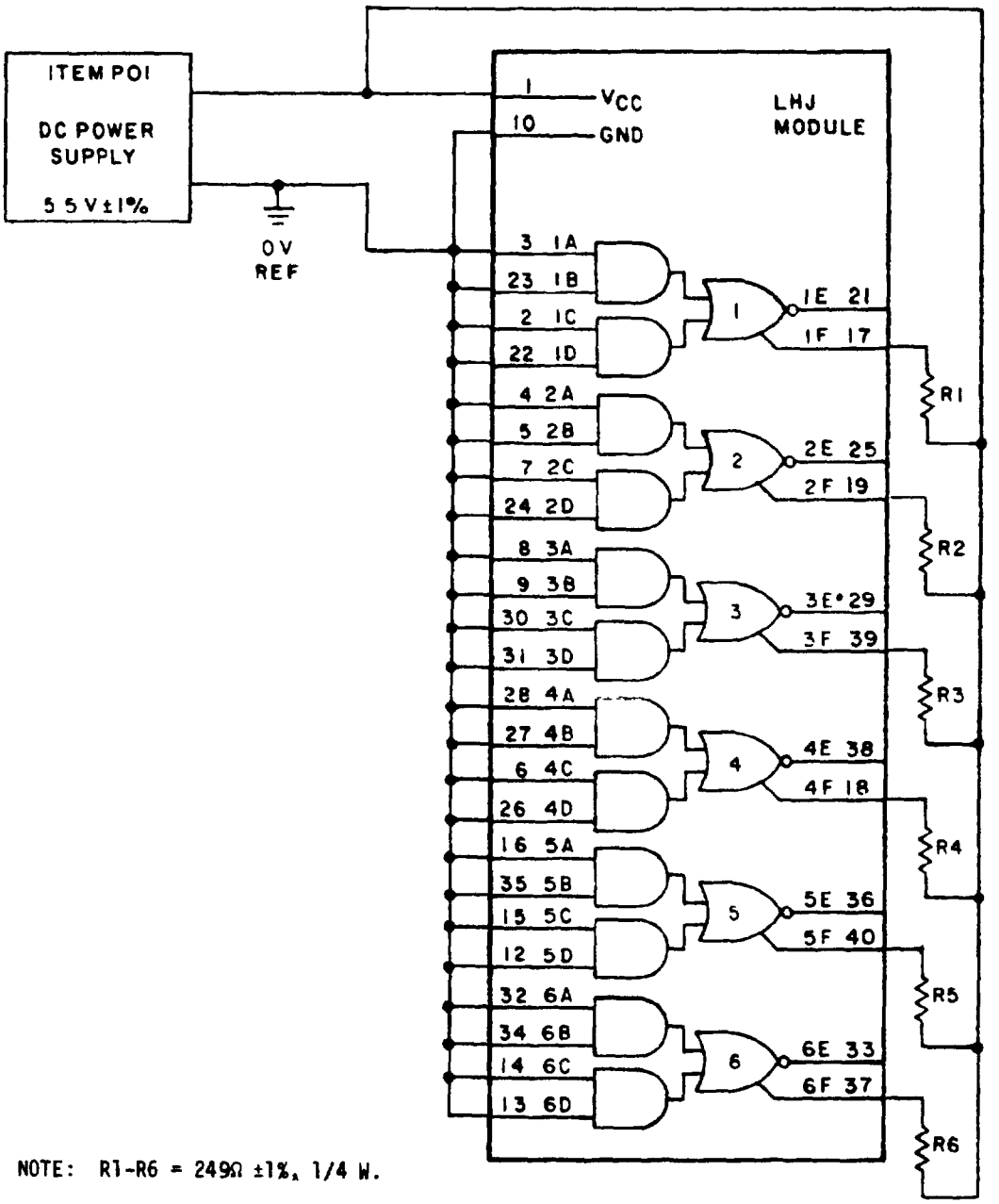


NOTES.
1. R1 = R3 = R5 = R6 = R8 = R10 = 249Ω ±1%, 1/4 W.
2. R2 = R4 = R7 = R9 = R13 = R14 = 277Ω ±1%, 1/4 W.
3. R11 = 1 kΩ ±5%, 1/2 W.
4. R12 = 51Ω ±5%, 1/2 W.
5. Pulse generator setup:

   Pulse width = 500 ±100 ns
   Frequency = 1 MHz ±20%
   $t_{TLH} = t_{THL}$ = 50 ±10 ns
   PA = 3.5 ±0.5 V.

FIGURE 2. Operating temperature and life test circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:49Z
Check the source to verify that this is the current version before use.



NOTE: R1-R6 = 249Ω ±1%, 1/4 W.

FIGURE 3. Component temperature test circuit

Source: https://assist.dla.mil -- Downloaded: 202640-20T15:49Z
Check the source to verify that this is the current version before use.

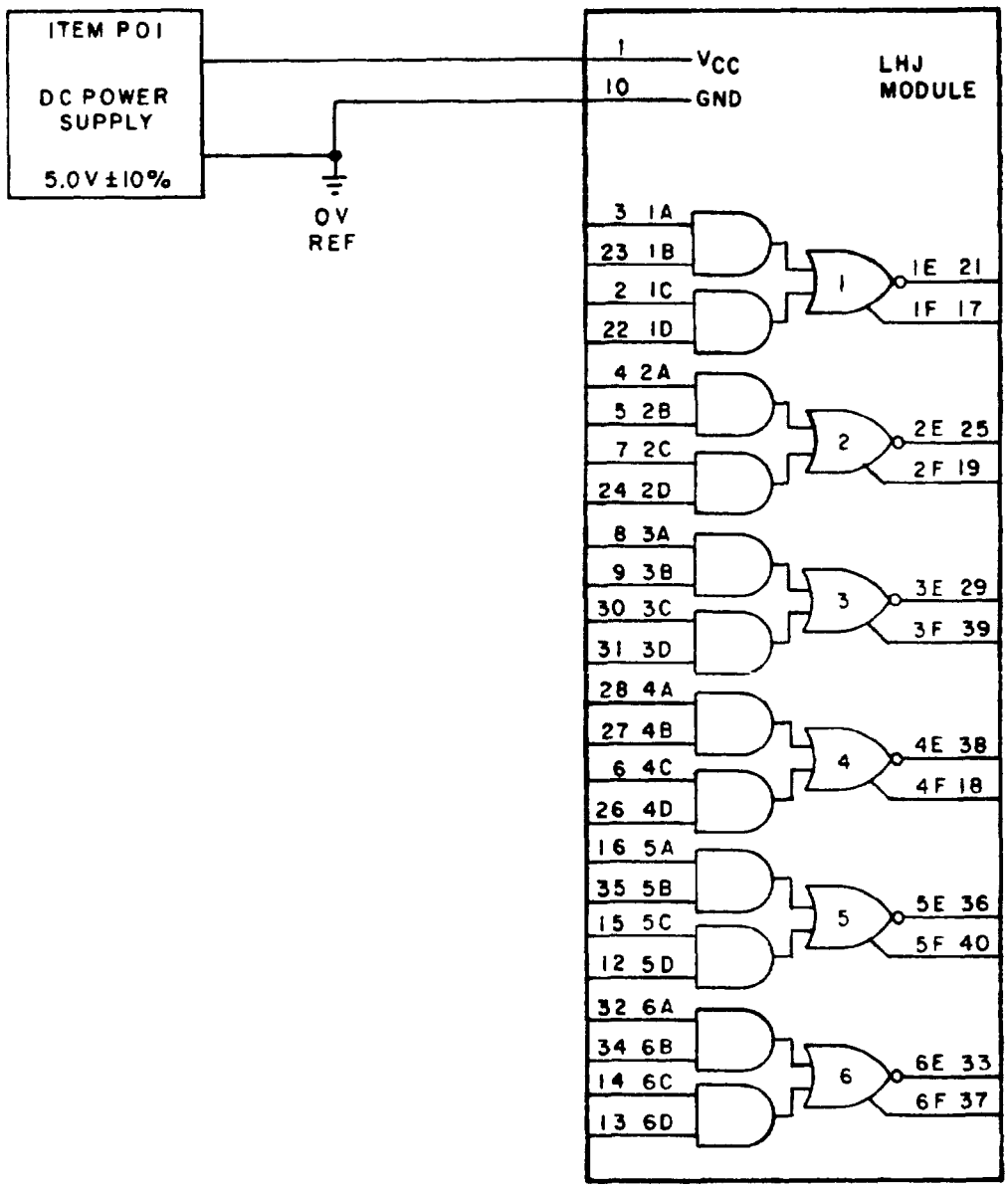


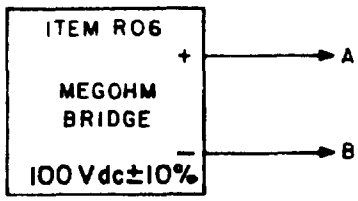
FIGURE 4. Vibration test circuit.

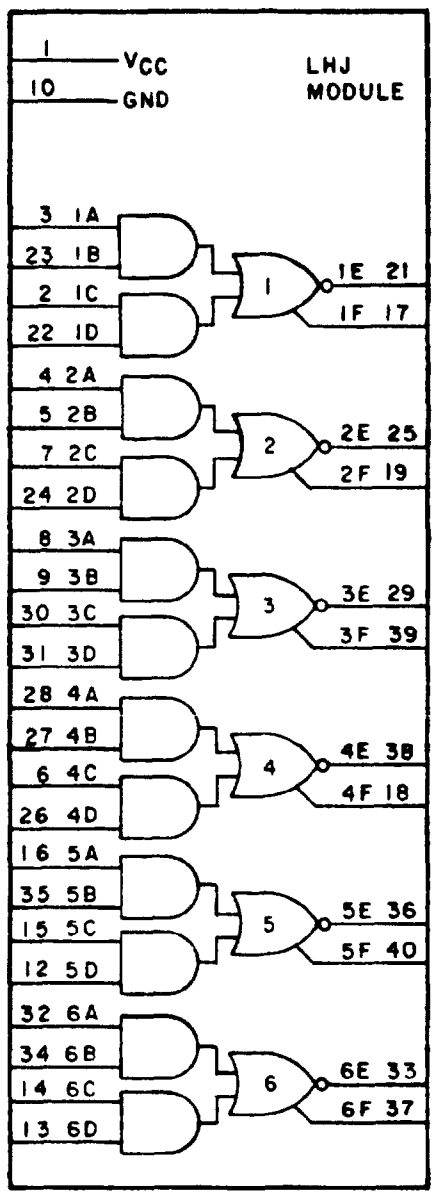
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:49Z
Check the source to verify that this is the current version before use.

TABLE III. Isolation test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A	Isolation	All pins	Frame	10	∞	MΩ	5

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:49Z
Check the source to verify that this is the current version before use.





WARNING·

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal

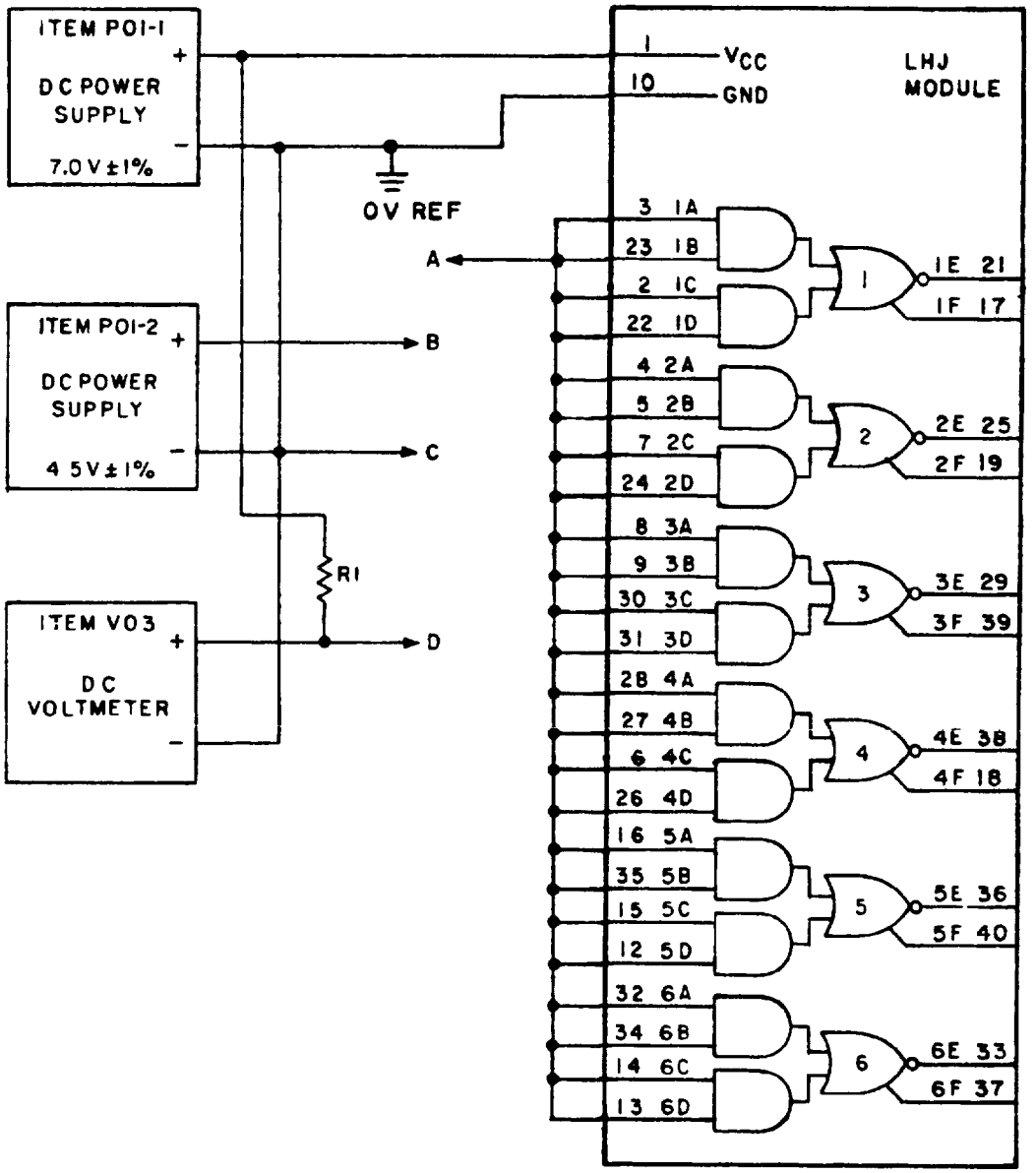
FIGURE 5. Isolation test circuit for table III.

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:49Z
Check the source to verify that this is the current version before use.

TABLE IV. V_{OUT} (low-loaded) test.

Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
B1	E1	V_{OUT} (low-loaded)	B	A	Open	21	0.01	1.0	V	6
B2	E2					25				
B3	E3					29				
B4	E4					38				
B5	E5					36				
B6	E6		B	A	Open	33				
B7	F1		C	Open	A	17				
B8	F2					19				
B9	F3					39				
B10	F4					18				
B11	F5					40				
B12	F6	V_{OUT} (low-loaded)	C	Open	A	37	0.01	1.0	V	6

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:49Z
Check the source to verify that this is the current version before use.



NOTE R1 = 249Ω ±1%, 1/4 W.

FIGURE 6. V_{OUT} (low-loaded) test circuit for table IV.

Source: https://assist.dla.mil -- Downloaded: 20260-20T15:49Z
Check the source to verify that this is the current version before use.

TABLE V. Input breakdown current test.

Test condition	Parameter	Connect: A to pin	Connect: B to pin	Initial limits 25° ± 5°C Ambient: Min	Initial limits 25° ± 5°C Ambient: Max	Unit	Test figure
C1	1A Input breakdown current	2,4,5,6,7,8,9,12,13,14,15, 16,22,23,24,26,27,28,30,31, 32,34,35	3	0.0	1.0	mA	7
C2	1B	2,3,4,5,6,7,8,9,12,13,14, 15,16,22,24,26,27,28,30,31, 32,34,35	23				
C3	1C	3,4,5,6,7,8,9,12,13,14,15, 16,22,23,24,26,27,28,30,31, 32,34,35	2				
C4	1D	2,3,4,5,6,7,8,9,12,13,14, 15,16,23,24,26,27,28,30,31, 32,34,35	22				
C5	2A	2,3,5,6,7,8,9,12,13,14,15, 16,22,23,24,26,27,28,30,31, 32,34,35	4				
C6	2B	2,3,4,6,7,8,9,12,13,14,15, 16,22,23,24,26,27,28,30,31, 32,34,35	5				
C7	2C	2,3,4,5,6,8,9,12,13,14,15, 16,22,23,24,26,27,28,30,31, 32,34,35	7				
C8	2D	2,3,4,5,6,7,8,9,12,13,14, 15,16,22,23,26,27,28,30,31, 32,34,35	24				
C9	3A Input breakdown current	2,3,4,5,6,7,9,12,13,14,15,16, 22,23,24,26,27,28,30,31, 32,34,35	8	0.0	1.0	mA	7

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T15:49Z
Check the source to verify that this is the current version before use.

TABLE V. Input breakdown current test - Continued.

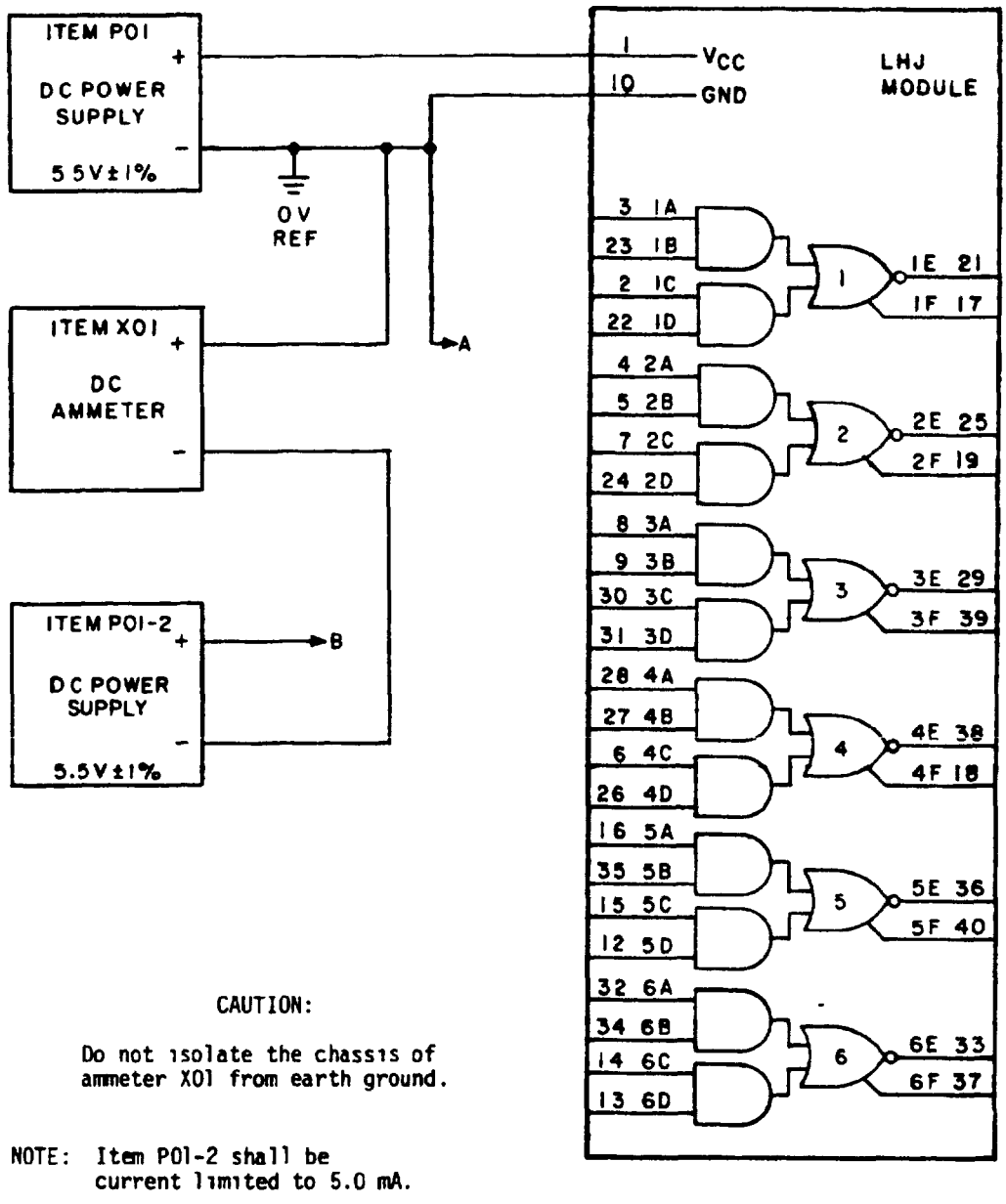
Test condition	Parameter	Connect: A to pin	Connect: B to pin	Initial limits 25° ± 5°C Ambient: Min	Initial limits 25° ± 5°C Ambient: Max	Unit	Test figure
C10	3B Input breakdown current	2,3,4,5,6,7,8,12,13,14,15, 16,22,23,24,26,27,28,30,31, 32,34,35	9	0.0	1.0	mA	7
C11	3C	2,3,4,5,6,7,8,9,12,13,14,15, 16,22,23,24,26,27,28,31, 32,34,35	30				
C12	3D	2,3,4,5,6,7,8,9,12,13,14, 15,16,22,23,24,26,27,28,30, 32,34,35	31				
C13	4A	2,3,4,5,6,7,8,9,12,13,14, 15,16,22,23,24,26,27,30,31, 32,34,35	28				
C14	4B	2,3,4,5,6,7,8,9,12,13,14, 15,16,22,23,24,26,28,30,31, 32,34,35	27				
C15	4C	2,3,4,5,7,8,9,12,13,14,15, 16,22,23,24,26,27,28,30,31, 32,34,35	6				
C16	4D	2,3,4,5,6,7,8,9,12,13,14, 15,16,22,23,24,27,28,30,31, 32,34,35	26				
C17	5A	2,3,4,5,6,7,8,9,12,13,14, 15,22,23,24,26,27,28,30,31, 32,34,35	16				
C18	5B Input breakdown current	2,3,4,5,6,7,8,9,12,13,14, 15,16,22,23,24,26,27,28,30, 31,32,34	35	0.0	1.0	mA	7

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:49Z
Check the source to verify that this is the current version before use.

TABLE V. Input breakdown current test - Continued

Test condition	Parameter	Connect		Initial limits 25° ± 5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C19	5C Input breakdown current	2,3,4,5,6,7,8,9,12,13,14, 16,22,23,24,26,27,28,30,31, 32,34,35	15	0.0	1.0	mA	7
C20	5D	2,3,4,5,6,7,8,9,13,14,15, 16,22,23,24,26,27,28,30,31, 32,34,35	12				
C21	6A	2,3,4,5,6,7,8,9,12,13,14, 15,16,22,23,24,26,27,28,30, 31,34,35	32				
C22	6B	2,3,4,5,6,7,8,9,12,13,14, 15,16,22,23,24,26,27,28,30, 31,32,35	34				
C23	6C	2,3,4,5,6,7,8,9,12,13,15, 16,22,23,24,26,27,28,30,31, 32,34,35	14				
C24	6D Input breakdown current	2,3,4,5,6,7,8,9,12,14,15, 16,22,23,24,26,27,28,30,31, 32,34,35	13	0.0	1.0	mA	7

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:49Z
Check the source to verify that this is the current version before use.



CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

NOTE: Item P01-2 shall be current limited to 5.0 mA.

FIGURE 7. Input breakdown current test circuit for table V.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:49Z
Check the source to verify that this is the current version before use.

TABLE VI. Output short circuit current test.

Test condition	Parameter		Connect			Initial limits 25° ± 5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
D1	1E	Output short circuit current	Open	2,3,4,5,6,7,8,9,12,13,14, 15,16,22,23,24,26,27,28, 30,31,32,34,35	21	-40	-100	mA	8
D2	2E			2,3,4,5,6,7,8,9,12,13,14, 15,16,22,23,24,26,27,28, 30,31,32,34,35	25				
D3	3E			2,3,4,5,6,7,8,9,12,13,14, 15,16,22,23,24,26,27,28, 30,31,32,34,35	29				
D4	4E			2,3,4,5,6,7,8,9,12,13,14, 15,16,22,23,24,26,27,28, 30,31,32,34,35	38				
D5	5E			2,3,4,5,6,7,8,9,12,13,14, 15,16,22,23,24,26,27,28, 30,31,32,34,35	36				
D6	6E		Open	2,3,4,5,6,7,8,9,12,13,14, 15,16,22,23,24,26,27,28, 30,31,32,34,35	33				
D7	1F		2,3, 22,23	4,5,6,7,8,9,12,13,14,15, 16,24,26,27,28,30,31,32, 34,35	17				
D8	2F		4,5, 7,24	2,3,6,8,9,12,13,14,15,16, 22,23,26,27,28,30,31,32, 34,35	19				
D9	3F		8,9, 30,31	2,3,4,5,6,7,12,13,14,15, 16,22,23,24,26,27,28,32, 34,35	39				
D10	4F	Output short circuit current	6,26, 27,28	2,3,4,5,7,8,9,12,13,14,15, 16,22,23,24,30,31,32,34,35	18	-40	-100	mA	8

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:49Z
Check the source to verify that this is the current version before use.

TABLE VI. Output short circuit current test - Continued.

Test condition	Parameter	Connect			Initial limits 25° ± 5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
D11	5F Output short circuit current	12,15, 16,35	2,3,4,5,6,7,8,9,13,14,22, 23,24,26,27,28,30,31,32,34	40	-40	-100	mA	8
D12	6F Output short circuit current	13,14, 32,34	2,3,4,5,6,7,8,9,12,15,16, 22,23,24,26,27,28,30,31,35	37	-40	-100	mA	8

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:49Z
Check the source to verify that this is the current version before use.

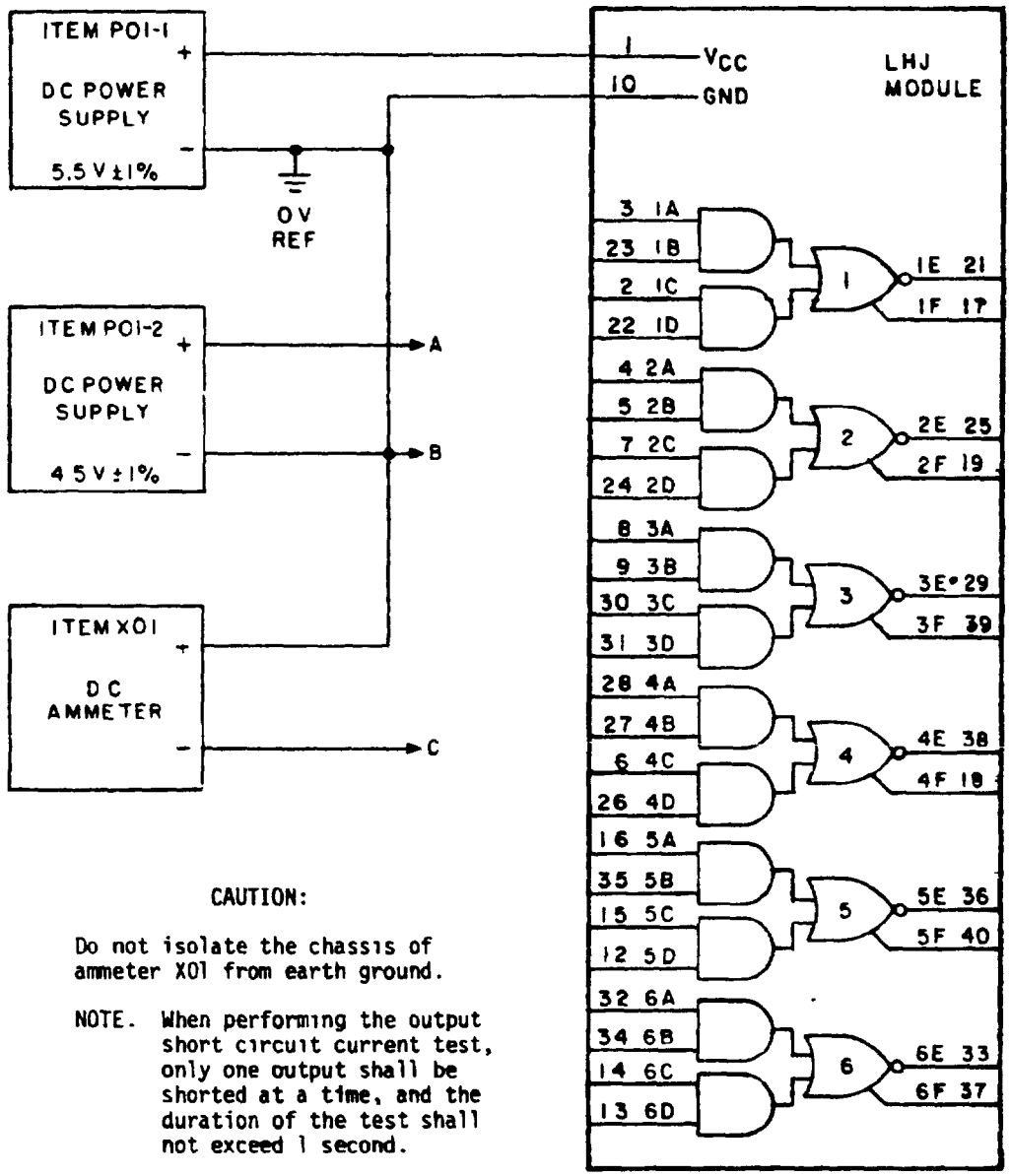


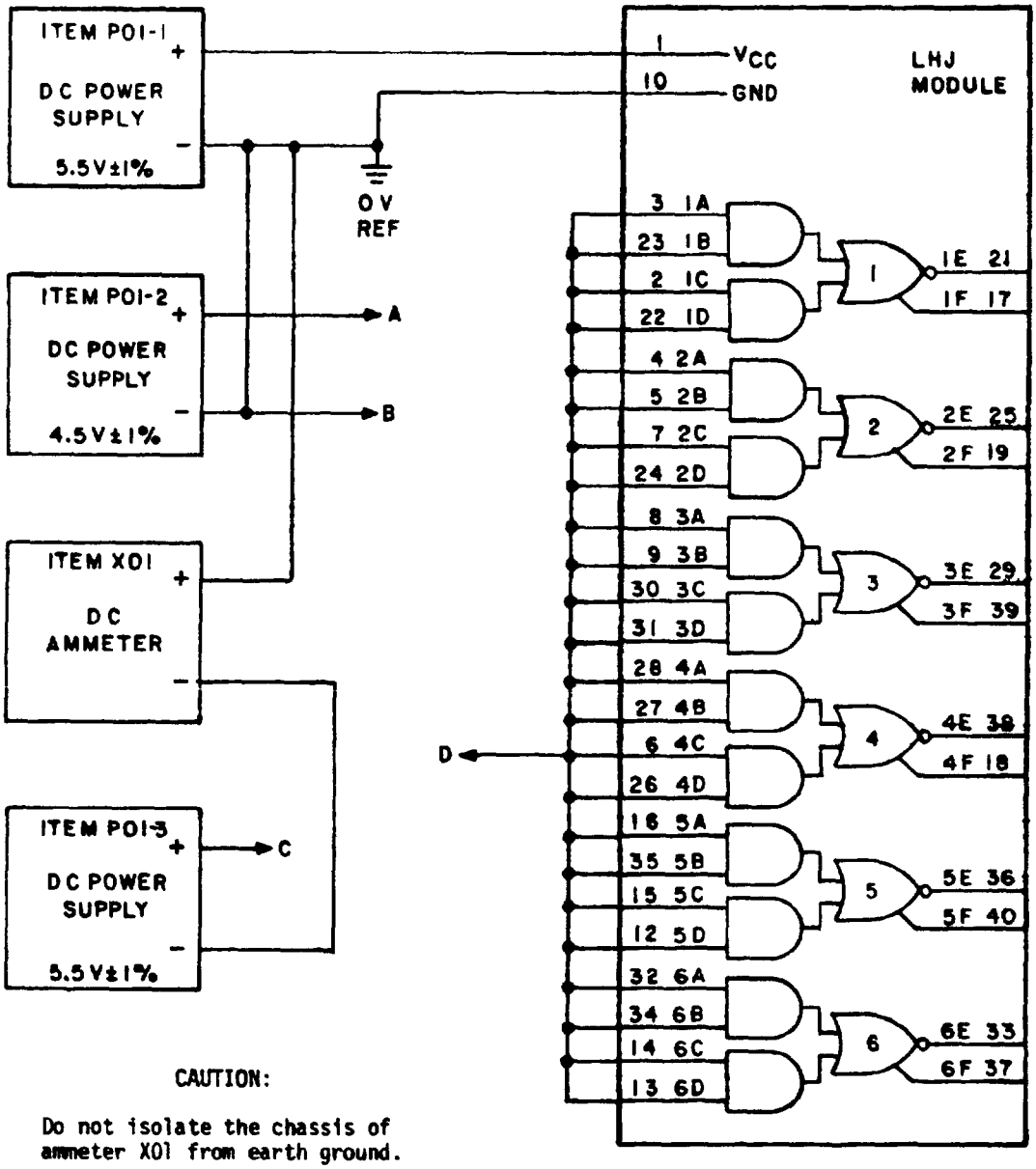
FIGURE 8 Output short circuit current test circuit for table VI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:49Z
Check the source to verify that this is the current version before use.

TABLE VII. Output leakage current test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
E1	1E Output leakage current	Open	D	21	0.0	4.0	mA	9
E2	2E			25				
E3	3E			29				
E4	4E			38				
E5	5E			36				
E6	6E	Open	D	33		4.0		
E7	1F	D	Open	17		1.0		
E8	2F			19				
E9	3F			39				
E10	4F			18				
E11	5F			40				
E12	6F Output leakage current	D	Open	37	0.0	1.0	mA	9

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:49Z
Check the source to verify that this is the current version before use.



CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

NOTE: Item P01-3 shall be current limited to 5 0 mA.

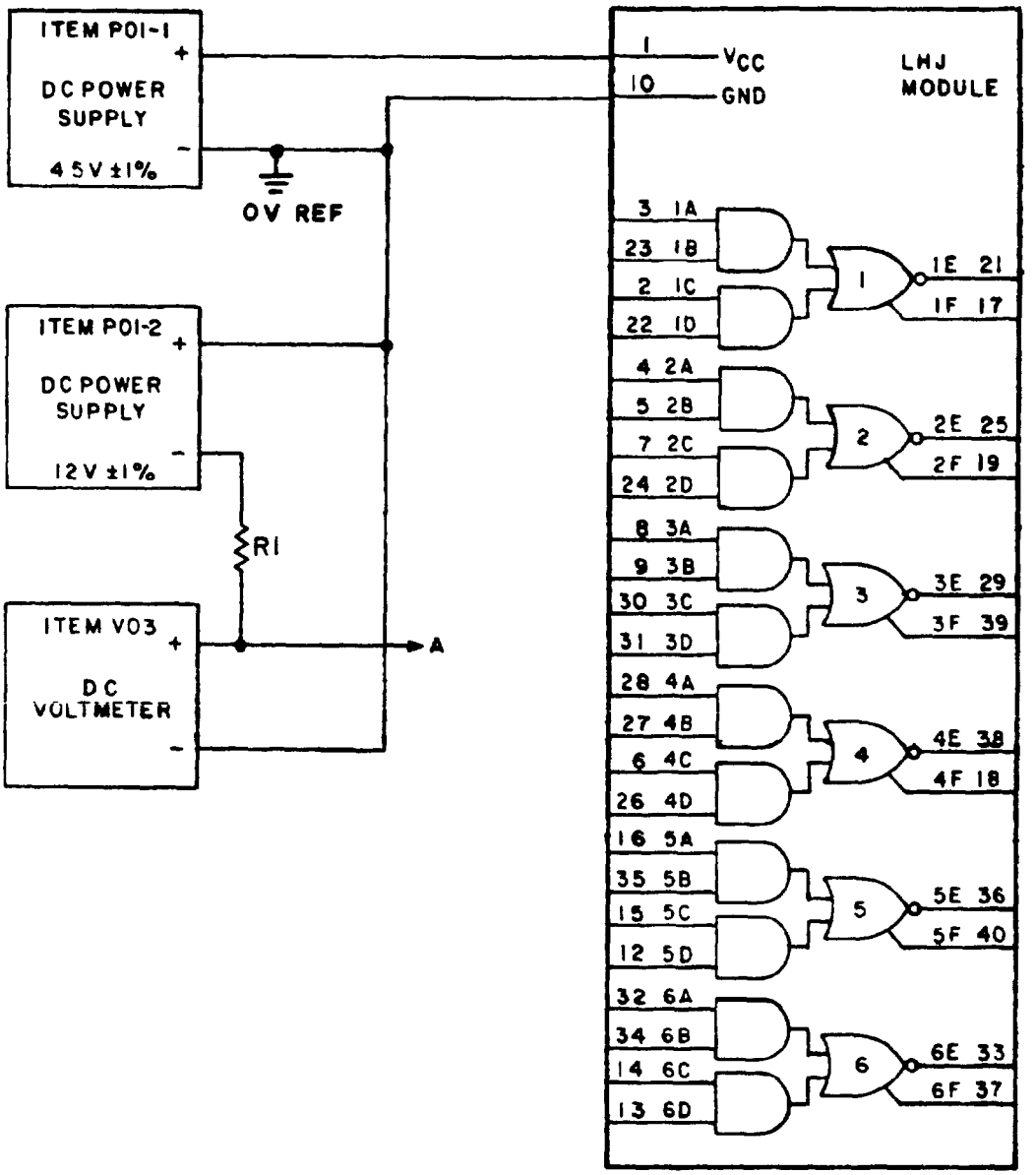
FIGURE 9 Output leakage current test circuit for table VII.

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:49Z
Check the source to verify that this is the current version before use.

TABLE VIII. Input clamp diode voltage test.

Test condition	Parameter		Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	Min	Max		
F1	1A	Input clamp diode voltage	3	-0.4	-1.2	V	10
F2	1B		23				
F3	1C		2				
F4	1D		22				
F5	2A		4				
F6	2B		5				
F7	2C		7				
F8	2D		24				
F9	3A		8				
F10	3B		9				
F11	3C		30				
F12	3D		31				
F13	4A		28				
F14	4B		27				
F15	4C		6				
F16	4D		26				
F17	5A		16				
F18	5B		35				
F19	5C		15				
F20	5D		12				
F21	6A		32				
F22	6B		34				
F23	6C		14				
F24	6D	Input clamp diode voltage	13	-0.4	-1.2	V	10

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:49Z
Check the source to verify that this is the current version before use.



NOTE. R1 = 590Ω ±1%, 1/4 W.

FIGURE 10 Input clamp diode voltage test circuit for table VIII.

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:49Z
Check the source to verify that this is the current version before use.

TABLE IX. Static output voltage (high-level and low-level) test.

Test condition	Parameter	Connect				Initial limits 25° ± 5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
G1	1E Static output voltage, high level	4,5,7,24, 8,9,30, 31,28,27, 6,26,16, 35,15,12, 32,34,14, 13	3,23,2, 22	21	17	2.5	4.5	V	11
G2	1E	5,24,9, 31,27,26, 35,12,34, 13	3,23,2, 22,4,7, 8,30,28, 6,16,15, 32,14	21	17				
G3	1F	3,23,22, 2,4,5,7, 24,8,9, 30,31,28, 27,6,26, 16,35,15, 12,32,34, 14,13	Open	17	21				
G4	1F	23,3,2, 22,5,24, 9,31,27, 26,35,12, 34,13	4,7,8,30, 28,6,16, 15,32,14	17	21				
G5	2E Static output voltage, high level	23,22,3, 2,8,9,30, 31,28,27, 6,26,16, 35,15,12, 32,34,14, 13	4,5,7, 24	25	19	2.5	4.5	V	11

Source: https://assist.dla.mil -- Downloaded: 2026/10-20T15:49Z
Check the source to verify that this is the current version before use.

TABLE IX. Static output voltage (high-level and low-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ± 5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
G6	2E Static output voltage, high level	2,23,9, 31,27,26, 35,12,34, 13	4,5,7, 24,3,22, 8,30,28, 6,16,15, 32,14	25	19	2.5	4.5	V	11
G7	2F	22,3,4,5, 7,2,24,8, 9,23,30, 31,28,27, 6,26,16, 35,15,12, 32,34,14, 13	Open	19	25				
G8	2F	3,22,4,5, 7,24,9, 31,27,26 35,12,34, 13	23,2,8, 30,28,6, 16,15,32, 14	19	25				
G9	3E	2,3,4,5, 23,7,24 28,27,6, 26,16,35, 34,22,14, 13	8,9,30, 31	29	39				
G10	3E Static output voltage, high level	23,22,5, 24,27,26, 35,12,34, 13	3,2,4,7, 8,9,30, 31,28,6, 16,15,32, 14	29	39	2.5	4.5	V	11

Source: https://assist.dla.mil -- Downloaded: 2026-0-20T15:49Z
Check the source to verify that this is the current version before use.

TABLE IX. Static output voltage (high-level and low-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ± 5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
G11	3F Static output voltage, high level	23,2,22, 3,4,5,7, 24,8,9, 30,31,28, 27,6,26, 16,35,15, 12,32,34, 14,13	Open	39	29	2 5	4.5	V	11
G12	3F	3,22,5, 24,8,9, 30,31,27, 26,35,12, 34,13	23,2,4, 7,28,6, 16,15,32, 14	39	29				
G13	4E	2,22,3, 23,4,5, 7,24,8, 9,30,31, 16,35,15, 12,32,34, 14,13	28,27,6, 26	38	18				
G14	4E	23,22,5, 24,9,31, 35,12,34, 13	27,28,6,26, 3,2,4,7,8, 30,16,15, 32,14	38	18				
G15	4F Static output voltage, high level	3,23,2,4, 5,7,24,8, 9,30,31, 22,28,27, 6,26,16, 35,15,12, 32,34,14, 13	Open	18	38	2 5	4.5	V	11

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:49Z
Check the source to verify that this is the current version before use.

TABLE IX. Static output voltage (high-level and low-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ± 5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
G16	4F Static output voltage, high level	23,22,5, 24,9,31, 28,27,6, 26,35,12, 34,13	3,2,4,7, 8,30,16, 15,32,14	18	38	2.5	4.5	V	11
G17	5E	3,23,2,4, 22,8,9,5, 30,31,28, 27,6,26, 32,34,7, 14,13,24	16,35,15, 12	36	40				
G18	5E	24,23,5, 22,9,31, 27,26,34, 13	4,7,3,2, 8,30,28, 6,32,14, 16,35,15, 12	36	40				
G19	5F	7,3,23,2, 22,8,9,4, 30,31,28, 27,6,26, 5,16,35, 15,12,32, 34,24,14, 13	Open	40	36				
G20	5F Static output voltage, high level	5,23,22, 9,24,31, 27,26,16, 35,15,12, 34,13	4,7,3,2, 8,30,28, 6,32,14	40	36	2.5	4.5	V	11

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:49Z
Check the source to verify that this is the current version before use.

TABLE IX. Static output voltage (high-level and low-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ± 5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
G21	6E Static output voltage, high level	5,24,3,4, 23,2,22, 7,8,9,30, 31,28,27, 6,26,16, 35,15,12	32,34,14, 13	33	37	2.5	4.5	V	11
G22	6E	5,7,23, 22,9,31, 27,26,35, 12	4,24,3,2, 8,30,28, 6,16,15, 32,34,14, 13	33	37				
G23	6F	4,3,23,2, 22,8,9,5, 30,31,28, 27,6,26, 7,16,35, 15,12,32, 34,24,14, 13	Open	37	33				
G24	6F Static output voltage, high level	4,24,23, 22,9,31, 27,26,35, 12,32,34, 14,15	3,5,7,2, 8,30,28, 6,16,15	37	33	2.5	4.5		
H1	1E Static output voltage, low level	4,5,24,7, 3,23,2, 22,8,9, 30,31,28, 27,6,35, 15,12,32, 34,14,13, 26,16	Open	21	17	0.0	0.5	V	11

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:49Z
Check the source to verify that this is the current version before use.

TABLE IX. Static output voltage (high-level and low-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ± 5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
H2	2E Static output voltage, low level	4,5,24,7, 3,23,2, 22,8,9, 30,31,28, 27,6,35, 15,12,32, 34,14,13, 26,16	Open	25	19	0.0	0.5	V	11
H3	3E	4,5,24,7, 3,23,2, 22,8,9, 30,31,28, 27,6,35, 15,12,32, 34,14,13, 26,16		29	39				
H4	4E	4,5,24,7, 3,23,2, 22,8,9, 30,31,28, 27,6,35, 15,12,32, 34,14,13, 26,16		38	18				
H5	5E Static output voltage, low level	4,5,3,23, 2,22,8,9, 30,31,28, 27,6,26, 16,35,15, 12,32,34, 7,14,13, 24	Open	36	40	0.0	0.5	V	11

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:49Z
Check the source to verify that this is the current version before use.

TABLE IX. Static output voltage (high-level and low-level) test - Continued.

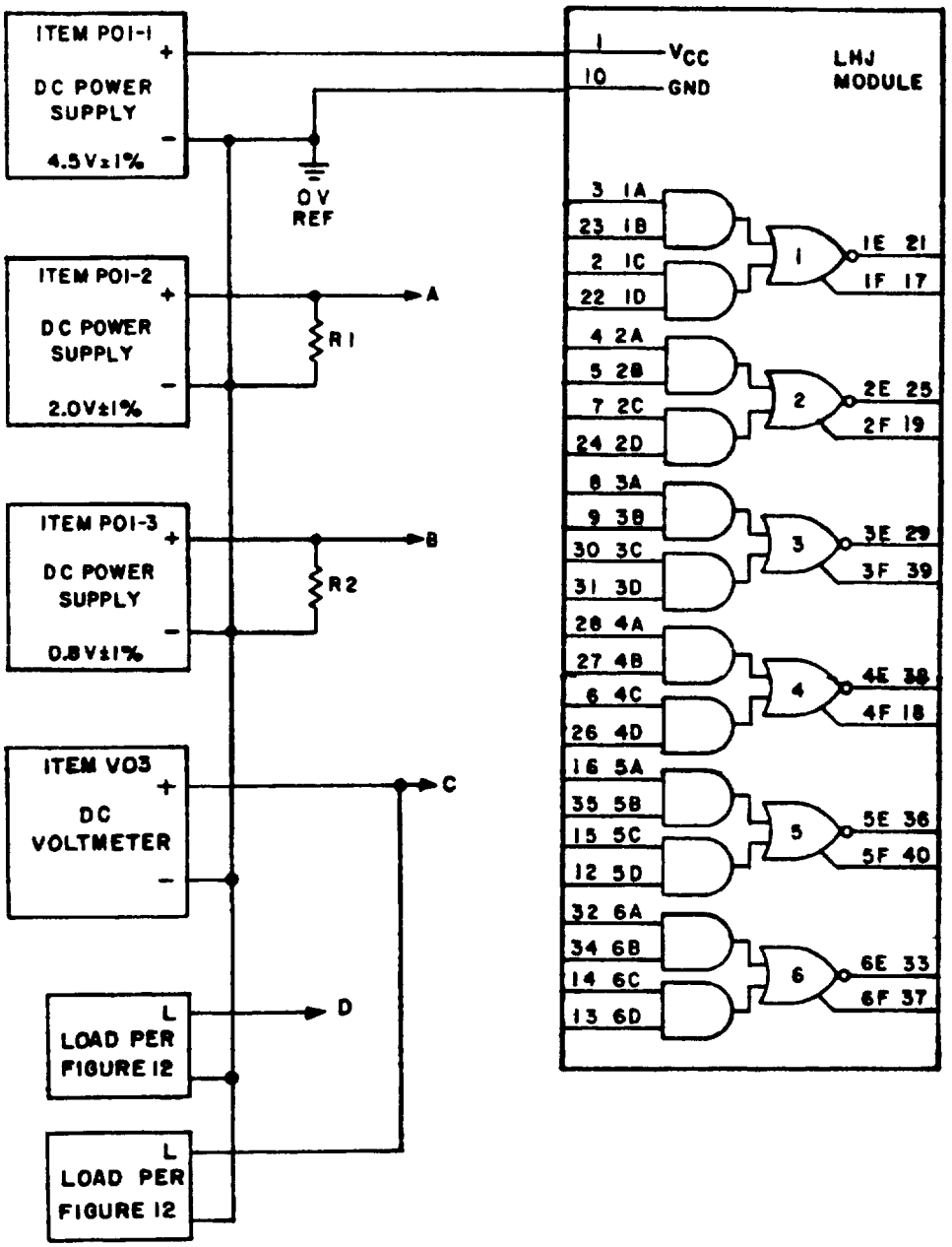
Test condition	Parameter	Connect				Initial limits 25° ± 5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
H6	6E Static output voltage, low level	4,5,24,3, 23,2,22, 8,9,30, 31,28,27, 6,20,16, 35,15,12, 32,34,14, 7,13	Open	33	37	0.0	0.5	V	11
H7	1F	Open	4,5,24,7, 3,23,2,22, 8,9,30,31, 28,27,6, 26,16,35, 15,12,32, 34,14,13	17	21				
H8	2F	Open	4,5,7,24,3, 23,2,22,8, 9,30,31,28, 27,6,26,16, 35,15,12,32, 34,14,13	19	25				
H9	3F Static output voltage, low level	Open	8,9,30,3, 23,31,2, 22,4,5,7, 24,28,27, 6,26,16, 35,15,12, 32,34,14, 13	39	29	0.0	0.5	V	11

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:49Z
Check the source to verify that this is the current version before use.

TABLE IX. Static output voltage (high-level and low-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ± 5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
H10	4F Static output voltage, low level	Open	8,9,30,3, 23,31,2, 22,4,5,7, 24,28,27, 6,26,16, 35,15,12, 32,34,14, 13	18	38	0.0	0.5	V	11
H11	5F Static output voltage, low level	Open	8,9,30,3, 23,31,2, 22,4,5,7, 24,28,27, 6,26,16, 35,15,12, 32,34,14, 13	40	36	0.0	0.5	V	11
H12	6F Static output voltage, low level	Open	8,9,30,3, 23,31,2, 22,4,5,7, 24,28,27, 6,26,16, 35,15,12, 32,34,14, 13	37	33	0.0	0.5	V	11

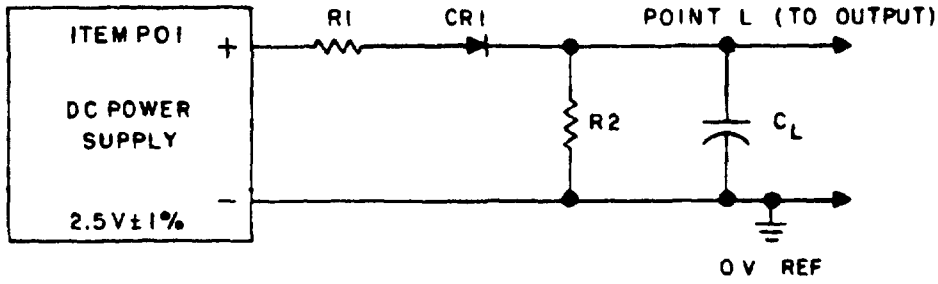
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:49Z
Check the source to verify that this is the current version before use.



NOTES
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. Both outputs of the circuit under test shall be loaded simultaneously per figure 12

* FIGURE 11. Static output voltage (high-level and low-level) test circuit for table IX.

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:49Z
Check the source to verify that this is the current version before use.



NOTES:
1. CR1 = 1N3600, 1N4150, or equivalent
2. RI = Select value for a current flow of 20 mA ±5% with point L held at 0.5 V for all F outputs. For all E outputs, select value for a current flow of 18 mA ±5% with point L held at 0.5 V.
3. R2 = 2.49 kΩ ±1%, 1/8 W for F outputs, 2.94 kΩ ±1%, 1/8 W for all E outputs.
4. Test type S:

   C_L = 50 ±5 pF (including probe and parasitic capacitance).

   Test type Q:

   C_L = 150 ±5 pF (including probe and parasitic capacitance) except for crosstalk test which will use 50 ±5 pF.
5. Capacitor C_L is not required for dc electrical test conditions.
6. Lead length from load circuit to module output pin under test is to be less than 1 inch (25.4 mm).

FIGURE 12. LHJ logic load circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:49Z
Check the source to verify that this is the current version before use.

TABLE X. Functional test.

Test cond	Parameter	Connect			Output logic states 1/												Test figure
		A to pin	B to pin	C to pin	E1	E2	E3	E4	E5	E6	F1	F2	F3	F4	F5	F6	
J1	Functional	Open	2,3,4,5,6, 7,8,9,12, 13,14,15, 16,22,23, 24,26,27, 28,30,31, 32,34,35	2/ and 3/	H	H	H	H	H	H	L	L	L	L	L	L	13
J2		12,13,22, 24,26,31,	2,3,4,5,6, 7,8,9,14, 15,16,23, 27,28,30, 32,34,35		H	H	H	H	H	H	L	L	L	L	L	L	
J3		2,6,7,14, 15,30	3,4,8,28, 16,32,23, 5,9,27,35, 34,22,24, 31,26,12, 13		H	H	H	H	H	H	L	L	L	L	L	L	
J4		2,7,6,30, 15,14,22, 24,31,26, 12,13	3,4,5,8,9, 28,16,32, 23,27,35, 34		L	L	L	L	L	L	H	H	H	H	H	H	
J5		23,5,9,27, 35,34	3,4,8,28, 16,32,2,7, 30,6,15, 14,22,24, 31,26,12, 13		H	H	H	H	H	H	L	L	L	L	L	L	
J6	Functional	23,5,9, 27,35,34, 22,24,31, 26,12,13	2,3,4,7,8, 6,14,15,16, 28,30,32	2/ and 3/	H	H	H	H	H	H	L	L	L	L	L	L	13

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026/10-20T15:49Z
Check the source to verify that this is the current version before use.

TABLE X. Functional test - Continued.

Test cond	Parameter	Connect			Output logic states 1/												Test figure
		A to pin	B to pin	C to pin	E1	E2	E3	E4	E5	E6	F1	F2	F3	F4	F5	F6	
J7	Functional	23,5,9, 27,35,34, 2,7,30,6, 15,14	3,4,8,28, 16,32,22, 24,31,26, 12,13	2/ and 3/	H	H	H	H	H	H	L	L	L	L	L	L	13
J8		23,5,9,27, 35,34,2,7, 30,6,15,14, 22,24,31, 26,12,13	3,4,16,28, 32,8		L	L	L	L	L	L	H	H	H	H	H	H	
J9		3,4,8,28, 16,32	23,5,9,27, 35,34,2,7, 30,6,15,14, 22,24,31, 26,12,13		H	H	H	H	H	H	L	L	L	L	L	L	
J10		3,4,8,28, 16,32,22, 24,31,26, 12,13	23,5,9,27, 35,34,2,7, 30,6,15,14		H	H	H	H	H	H	L	L	L	L	L	L	
J11		3,4,8,28, 16,32,2, 7,30,6, 15,14	23,5,9,27, 35,34,22, 24,31,26, 12,13		H	H	H	H	H	H	L	L	L	L	L	L	
J12		3,4,8,28, 16,32,2,7, 30,6,15, 14,22,24, 31,26,12, 13	23,5,9, 35,34		L	L	L	L	L	L	H	H	H	H	H	H	
J13	Functional	3,4,8,28, 16,32,23, 5,9,27, 35,34	2,7,30,6, 15,14,22, 24,21,26, 12,13	2/ and 3/	L	L	L	L	L	L	H	H	H	H	H	H	13

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:49Z
Check the source to verify that this is the current version before use.

TABLE X. Functional test - Continued.

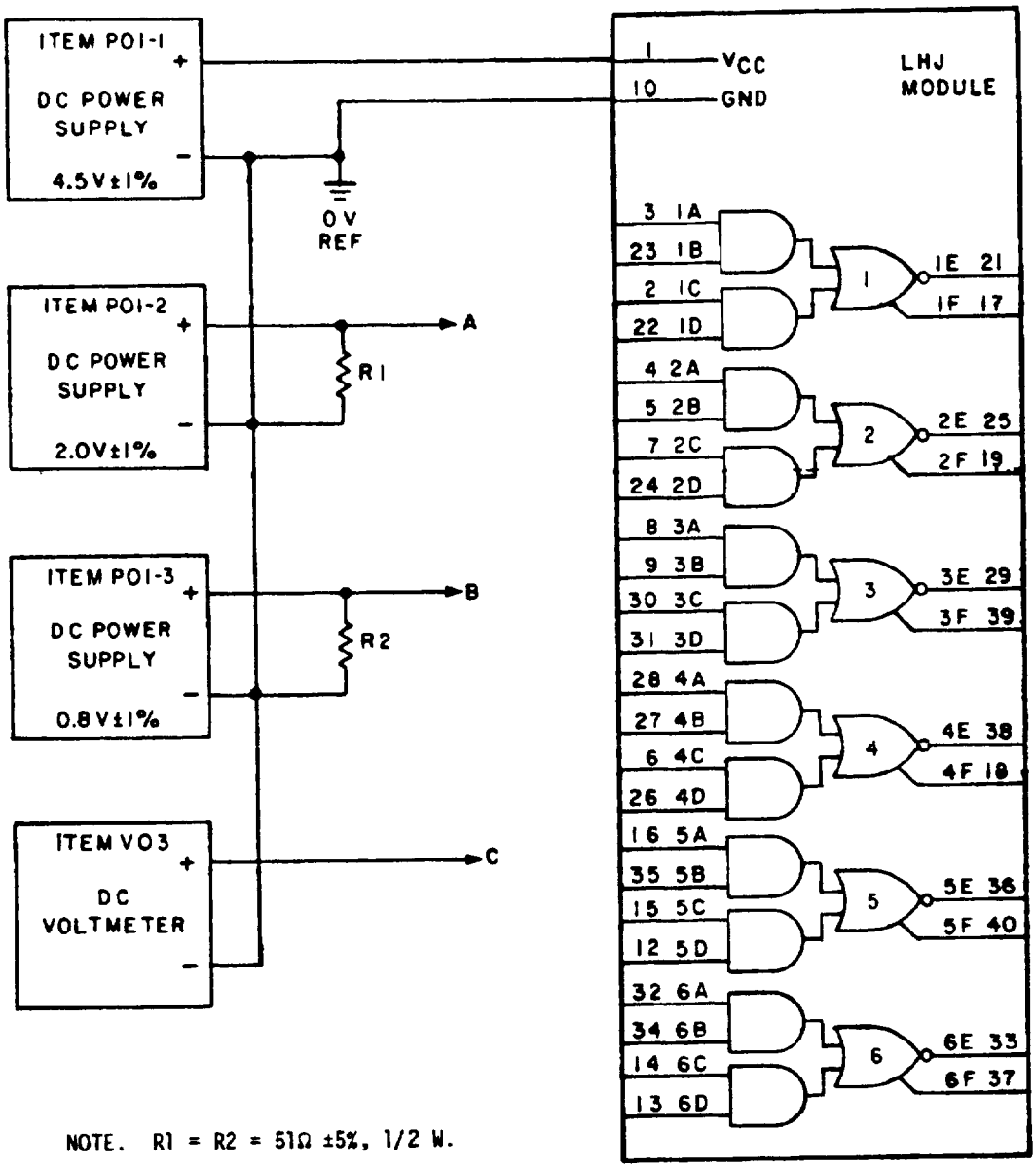
Test cond	Parameter	Connect			Output logic states 1/												Test figure
		A to pin	B to pin	C to pin	E1	E2	E3	E4	E5	E6	F1	F2	F3	F4	F5	F6	
J14	Functional	3,4,8,28, 16,32,23, 5,9,27, 35,34,22, 24,31,26, 12,13	2,7,30,6, 15,14	2/ and 3/	L	L	L	L	L	L	H	H	H	H	H	H	13
J15	Functional	3,4,8,28, 16,32,23, 5,9,27, 35,34,2, 7,30,6, 15,14	22,24,31, 26,12,13	2/ and 3/	L	L	L	L	L	L	H	H	H	H	H	H	13
J16	Functional	3,4,8,28, 16,32,23, 5,9,27, 35,34,2,7, 30,6,15, 14,22,24, 31,26,12, 13	Open	2/ and 3/	L	L	L	L	L	L	H	H	H	H	H	H	13

1/ Output logic states - L = Low-level voltage = 0.0 to 0.5 V
H = High-level voltage = 2.5 to 4.5 V

2/ For test conditions J1-J16 verification of the output logic states in 1/ is obtained by connecting item (V03), point C, to each of the output pins in succession after input conditions have been accomplished.

3/ E1, E2, E3, E4, E5, E6, F1, F2, F3, F4, F5, and F6 output logic states are measured at output pins 21, 25, 29, 38, 36, 33, 17, 19, 39, 18, 40, and 37, respectively.

Source: https://assist.dla.mil -- Downloaded: 2026/0-20T15:49Z
Check the source to verify that this is the current version before use.



NOTE. R1 = R2 = 51Ω ±5%, 1/2 W.

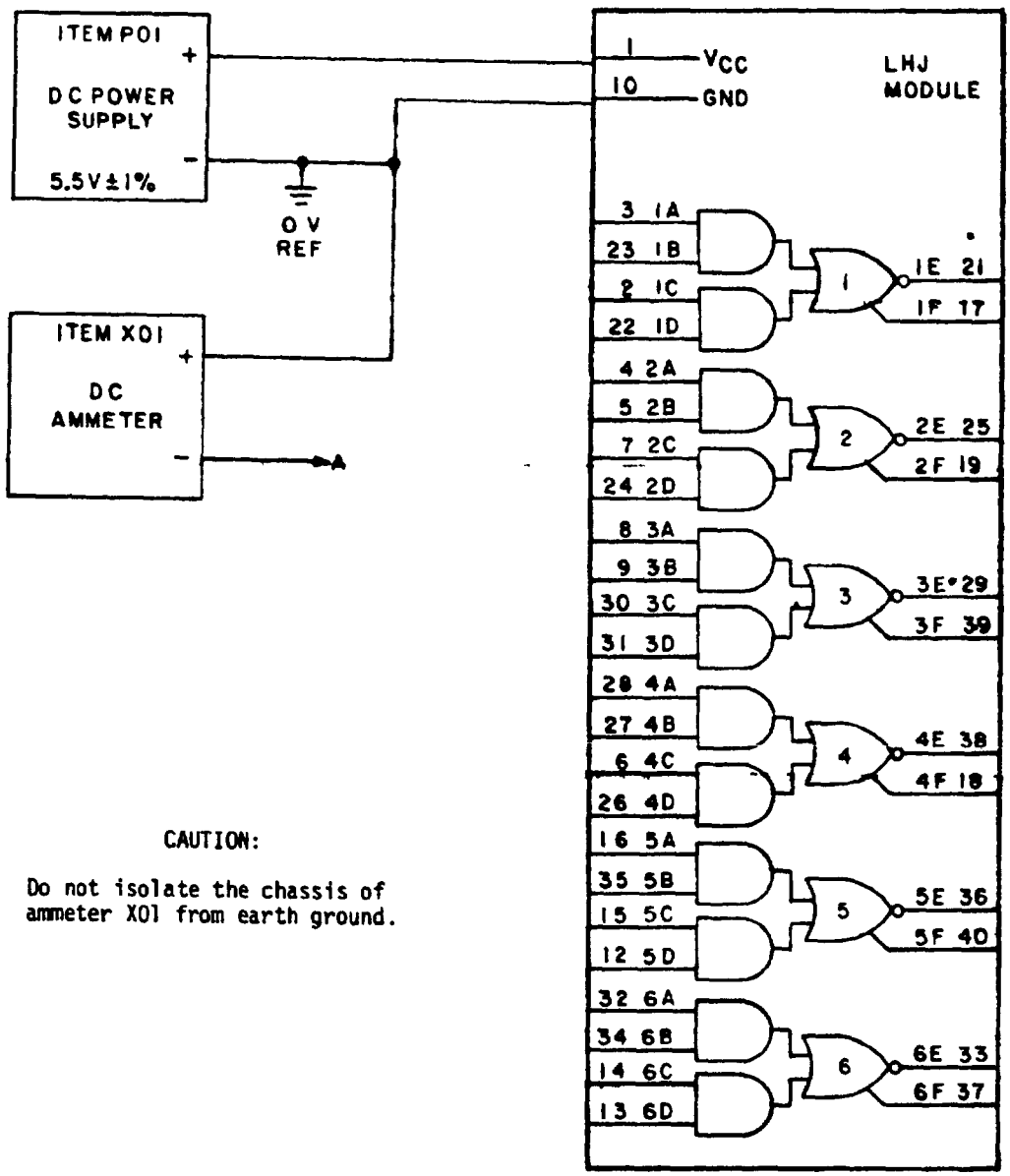
FIGURE 13. Functional test circuit for table X.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:49Z
Check the source to verify that this is the current version before use.

TABLE XI. Input current (low-level) test.

Test condition	Parameter		Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	Min	Max		
K1	1A	Input current, low level	3	-0.1	-2.2	mA	14
K2	1B		23				
K3	1C		2				
K4	1D		22				
K5	2A		4				
K6	2B		5				
K7	2C		7				
K8	2D		24				
K9	3A		8				
K10	3B		9				
K11	3C		30				
K12	3D		31				
K13	4A		28				
K14	4B		27				
K15	4C		6				
K16	4D		26				
K17	5A		16				
K18	5B		35				
K19	5C		15				
K20	5D		12				
K21	6A		32				
K22	6B		34				
K23	6C		14				
K24	6D	Input current, low level	13	-0.1	-2.2	mA	14

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:49Z
Check the source to verify that this is the current version before use.



* FIGURE 14. Input current (low-level) test circuit for table XI.

Source: https://assist.dla.mil -- Downloaded: 20260-20T15:49Z
Check the source to verify that this is the current version before use.

TABLE XII. Input current (high-level) test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
L1	1A Input current, high level	2,22,4,5,7, 24,8,9,30,31, 28,27,6,26, 16,35,15,12, 32,34,14,13, 23	3	0	50	µA	15
L2	1B	2,22,4,5,7, 24,8,9,30,31, 28,27,6,26, 16,35,15,12, 32,34,14,13, 3	23				
L3	1C	3,23,4,5,7, 24,8,9,30,31, 28,27,6,26, 16,35,15,12, 32,34,14,13, 22	2				
L4	1D	3,23,4,5,7, 24,8,9,30,31, 28,27,6,26, 16,35,15,12, 32,34,14,13, 2	22				
L5	2A Input current, high level	3,23,2,22,7, 24,8,9,30,31, 28,27,6,26, 16,35,15,12, 32,34,14,13, 5	4	0	50	µA	15

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:49Z
Check the source to verify that this is the current version before use.

TABLE XII. Input current (high-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
L6	2B Input current, high level	3,23,1,22,7, 24,8,9,30,31, 28,27,6,26, 16,35,15,12, 32,34,14,13,4	5	0	50	μA	15
L7	2C	3,23,1,22,4, 5,8,9,30,31, 28,27,6,26, 16,35,15,12, 32,34,14,13, 24	7				
L8	2D	3,23,2,22,4, 5,8,9,30,31, 28,27,6,26, 16,35,15,12, 32,34,14,13,7	24				
L9	3A	3,23,2,22,4, 5,7,24,30,31, 28,27,6,26, 16,35,15,12, 32,34,14,13,9	8				
L10	3B	3,23,2,22,4, 5,7,24,30,31, 28,27,6,26, 16,35,15,12, 32,34,14,13,8	9				
L11	3C Input current, high level	3,23,2,22,4, 5,7,24,28,27, 6,26,16,35, 15,12,32,34, 14,13,31	30	0	50	uA	15

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:49Z
Check the source to verify that this is the current version before use.

TABLE XII. Input current (high-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
L12	3D Input current, high level	3,23,2,22,4, 5,7,24,28,27, 6,26,16,35, 15,12,32,34, 14,13,30	31	0	50	µA	15
L13	4A	3,23,2,22,4, 5,7,24,30,31, 6,26,16,35, 15,12,32,34, 14,13,27	28				
L14	4B	3,23,2,22,4, 5,7,24,30,31, 6,26,16,35, 15,12,32,34, 14,13,28	27				
L15	4C	3,23,2,22,4, 5,7,24,30,31, 27,28,16,35, 15,12,32,34, 14,13,26	6				
L16	4D	3,23,2,22,4, 5,7,24,30,31, 27,28,16,35, 15,12,32,34, 14,13,6	26				
L17	5A Input current, high level	3,23,2,22,4, 5,7,24,8,9, 30,31,28,27, 6,26,15,12, 32,34,14,13, 35	16	0	50	µA	15

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:49Z
Check the source to verify that this is the current version before use.

TABLE XII. Input current (high-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
L18	5B Input current, high level	3,23,2,22,4, 5,7,24,8,9, 30,31,28,27, 6,26,15,12, 32,34,14,13, 16	35	0	50	µA	15
L19	5C	3,23,2,22,4, 5,7,24,8,9, 30,31,28,27, 6,26,16,35, 32,34,14,13, 12	15				
L20	5D	3,23,2,22,4, 5,7,24,8,9, 30,31,28,27, 6,26,16,35, 32,34,14,13, 15	12				
L21	6A	3,23,2,22,4, 5,7,24,8,9, 30,31,28,27, 6,26,16,35, 15,12,14,13, 34	32				
L22	6B Input current, high level	3,23,2,22,4, 5,7,24,8,9, 30,31,28,27, 6,26,16,35, 15,12,14,13, 32	34	0	50	µA	15

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T15:49Z
Check the source to verify that this is the current version before use.

TABLE XII. Input current (high-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
L23	6C Input current, high level	3,23,2,22,4, 5,7,24,8,9, 30,31,28,27, 6,26,16,35, 15,12,32,34, 13	14	0	50	µA	15
L24	6D Input current, high level	3,23,2,22,4, 5,7,24,8,9, 30,31,28,27, 6,26,16,35, 15,12,32,34, 14	13	0	50	µA	15

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:49Z
Check the source to verify that this is the current version before use.

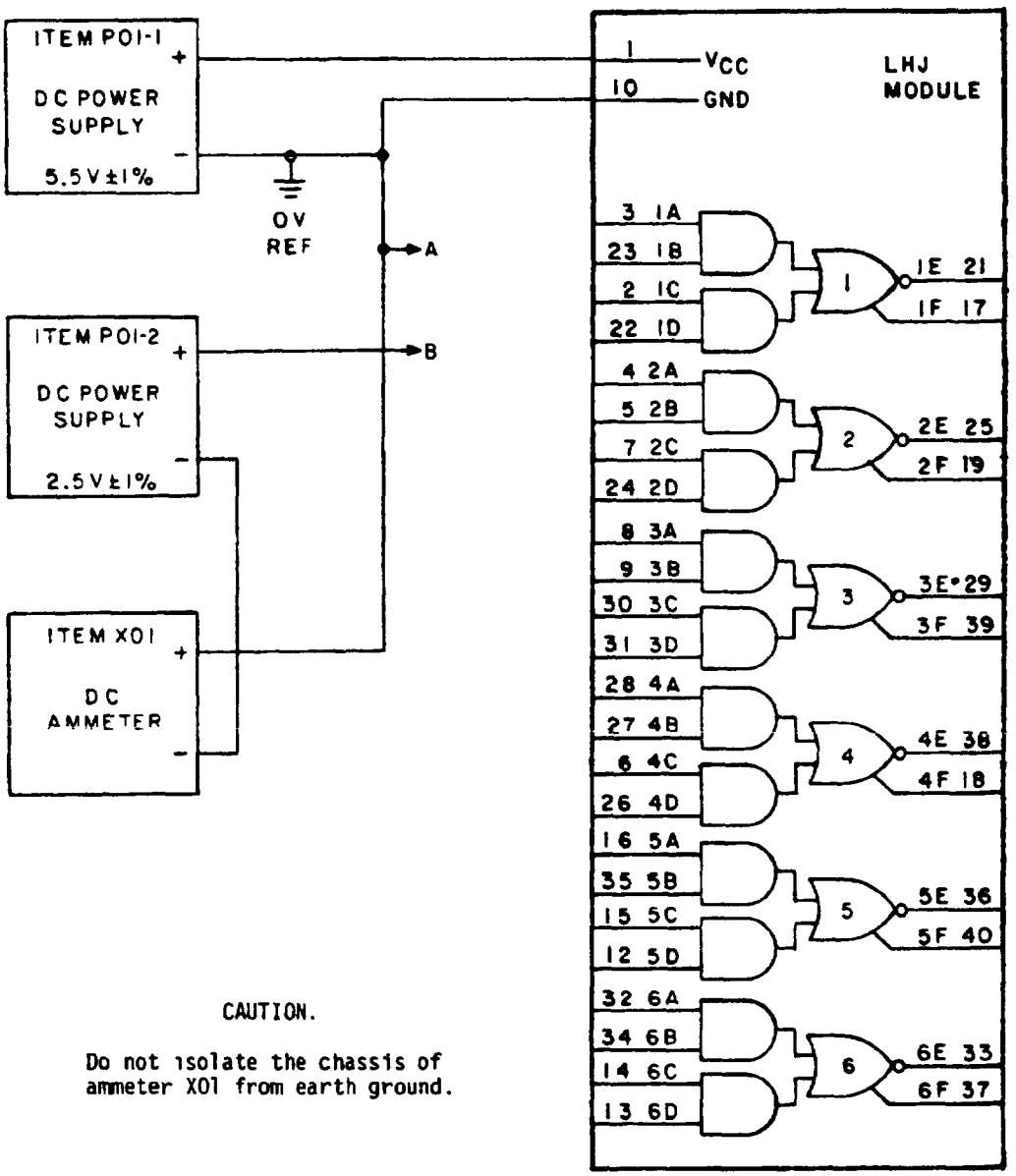


FIGURE 15. Input current (high-level) test circuit for table XII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:49Z
Check the source to verify that this is the current version before use.

TABLE XIII. Power supply current (low-level outputs) test.

Test condition	Parameter	Initial limits 25° ±5°C Ambient		Unit	Test figure
		Min	Max		
M	Power supply current, low level outputs	40	120	mA	16

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:49Z
Check the source to verify that this is the current version before use.

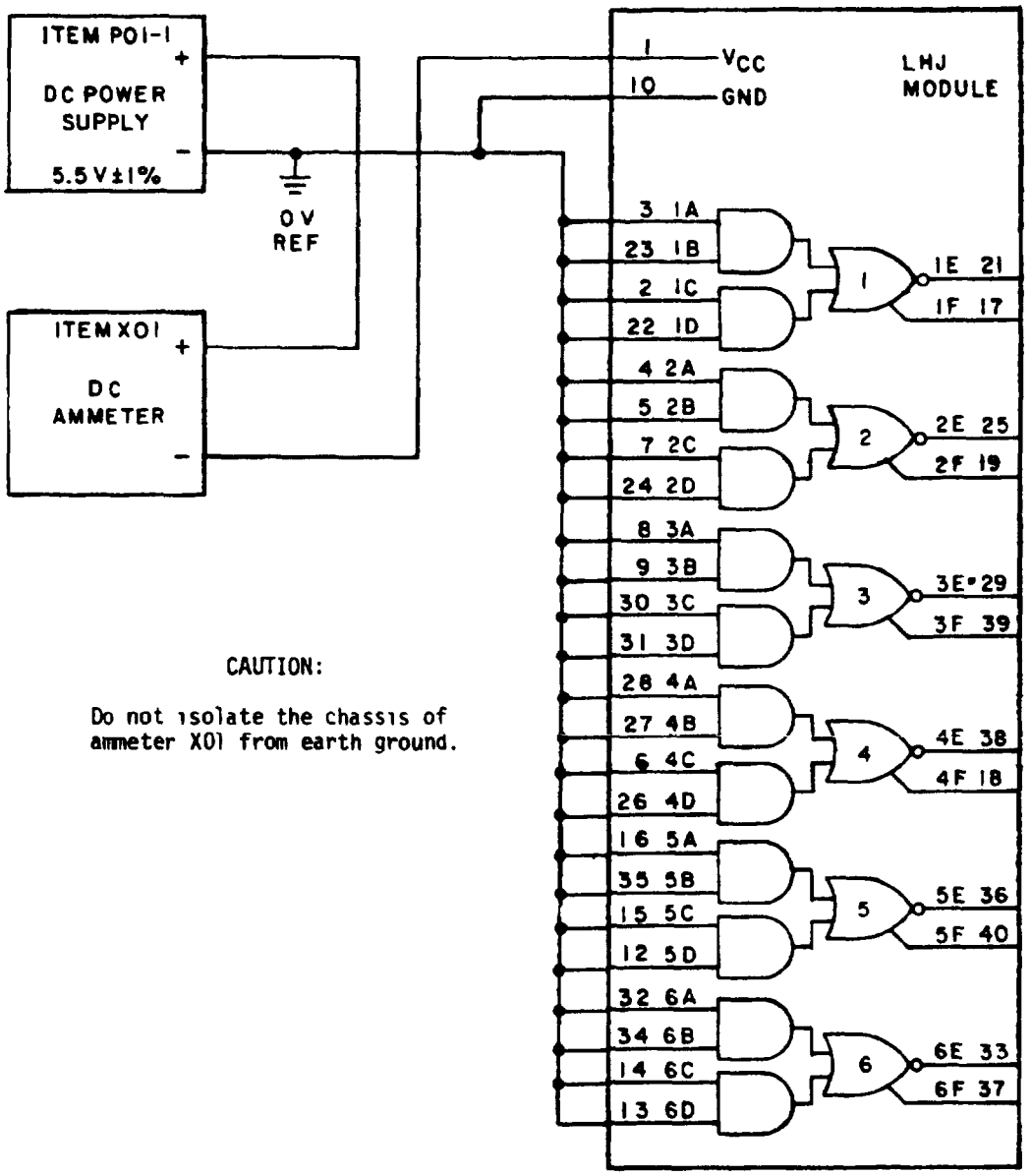


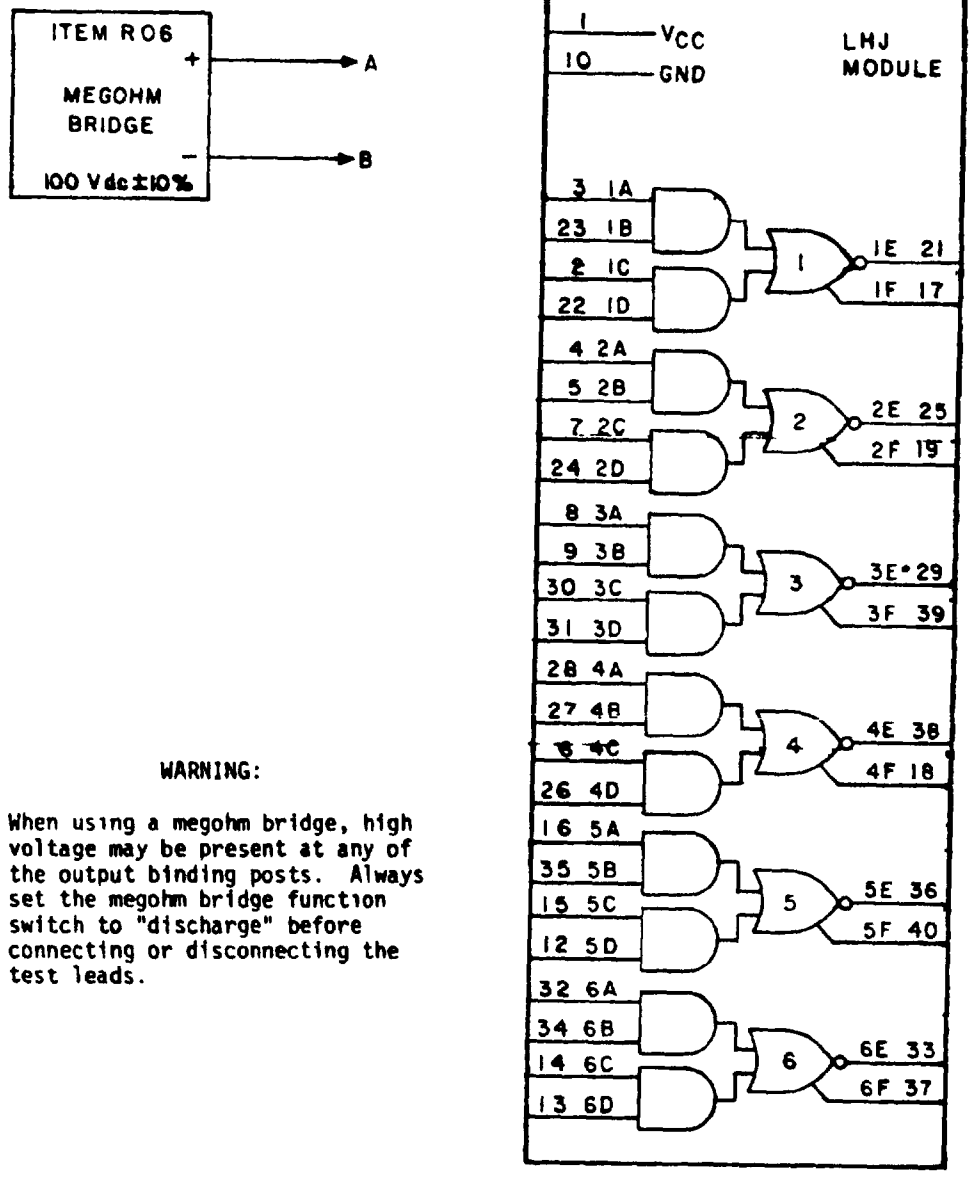
FIGURE 16. Power supply current (low-level outputs) test circuit for table XIII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:49Z
Check the source to verify that this is the current version before use.

TABLE XIV. Unused pin isolation.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
N1	Unused pin isolation, pin 11	All pins except 11 and 20	11	10.0	∞	MΩ	17
N2	Unused pin isolation, pin 20	All pins except 11 and 20	20	10.0	∞	MΩ	17

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:49Z
Check the source to verify that this is the current version before use.



WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

FIGURE 17. Unused pin isolation test circuit for table XIV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:49Z
Check the source to verify that this is the current version before use.

* TABLE XV. Transition time (t_{TLH} and t_{THL}) test.

Test condition	Parameter		Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Initial limits 25° +5°C Ambient Min	Initial limits 25° +5°C Ambient Max	Unit	Test figure
P1	1E	Transition time (t_{TLH}), C_L = 50 pF	23	2,22	3	21	2.0	17.0	ns	18,19
P2	1F		23	2,22	3	17		21.0		
P3	2E		5	7,24	4	25		17.0		
P4	2F		5	7,24	4	19		21.0		
P5	3E		9	30,31	8	29		17.0		
P6	3F		9	30,31	8	39		21.0		
P7	4E		27	6,26	28	38		17.0		
P8	4F		27	6,26	28	18		21.0		
P9	5E		35	15,12	16	36		17.0		
P10	5F		35	15,12	16	40		21.0		
P11	6E		34	14,13	32	33		17.0		
P12	6F	Transition time (t_{TLH}), C_L = 50 pF	34	14,13	32	37		21.0		
P13	1E	Transition time (t_{TLH}), C_L = 150 pF	23	2,22	3	21		32.0		
P14	1F		23	2,22	3	17				
P15	2E		5	7,24	4	25				
P16	2F	Transition time (t_{TLH}), C_L = 150 pF	5	7,24	4	19	2.0	32.0	ns	18,19

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:49Z
Check the source to verify that this is the current version before use.

* TABLE XV. Transition time (t_{TLH} and t_{THL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
P17	3E	Transition time (t_{TLH}), C_L = 150 pF	9	30,31	8	29	2.0	32.0	ns	18,19
P18	3F	↓	9	30,31	8	39	↓	↓	↓	↓
P19	4E		27	6,26	28	38				
P20	4F		27	6,26	28	18				
P21	5E		35	15,12	16	36				
P22	5F		35	15,12	16	40				
P23	6E		34	14,13	32	33		↓		
P24	6F	Transition time (t_{TLH}), C_L = 150 pF	34	14,13	32	37		32.0		
Q1	1E	Transition time (t_{THL}), C_L = 50 pF	23	2,22	3	21		9.0		
Q2	1F	↓	23	2,22	3	17		↓		
Q3	2E		5	7,24	4	25				
Q4	2F		5	7,24	4	19				
Q5	3E		9	30,31	8	29				
Q6	3F		9	30,31	8	39				
Q7	4E		27	6,26	28	38	↓	↓	↓	↓
Q8	4F	Transition time (t_{THL}), C_L = 50 pF	27	6,26	28	18	2.0	9.0	ns	18,19

Source: https://assist.dla.mil -- Downloaded: 20260-20T15:49Z
Check the source to verify that this is the current version before use.

* TABLE XV. Transition time (t_{TLH} and t_{THL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Q9	5E	Transition time (t_{THL}), C_L = 50 pF	35	15,12	16	36	2.0	9.0	ns	18,19
Q10	5F	↓	35	15,12	16	40	↓	↓	↓	↓
Q11	6E	↓	34	14,13	32	33	↓	↓	↓	↓
Q12	6F	Transition time (t_{THL}), C_L = 50 pF	34	14,13	32	37	↓	9.0	↓	↓
Q13	1E	Transition time (t_{THL}), C_L = 150 pF	23	2,22	3	21	↓	19.0	↓	↓
Q14	1F	↓	23	2,22	3	17	↓	↓	↓	↓
Q15	2E	↓	5	7,24	4	25	↓	↓	↓	↓
Q16	2F	↓	5	7,24	4	19	↓	↓	↓	↓
Q17	3E	↓	9	30,31	8	29	↓	↓	↓	↓
Q18	3F	↓	9	30,31	8	39	↓	↓	↓	↓
Q19	4E	↓	27	6,26	28	38	↓	↓	↓	↓
Q20	4F	↓	27	6,26	28	18	↓	↓	↓	↓
Q21	5E	↓	35	15,12	16	36	↓	↓	↓	↓
Q22	5F	↓	35	15,12	16	40	↓	↓	↓	↓
Q23	6E	↓	34	14,13	32	33	↓	↓	↓	↓
Q24	6F	Transition time (t_{THL}), C_L = 150 pF	34	14,13	32	37	2.0	19.0	ns	18,19

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:49Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026/10-20T15:49Z
Check the source to verify that this is the current version before use.

* TABLE XVI. Propagation delay time (t_{PLH} and t_{PHL}) test.

Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
R1	1A to 1E	Propagation delay time (t_{PLH}), C_L = 50 pF	23	2,22	3	21	3.5	15.0	ns	18,20
R2	1B to 1E	↓	3	2,22	23	21	↓	↓	↓	↓
R3	1C to 1E		22	3,23	2	21				
R4	1D to 1E		2	3,23	22	21				
R5	2A to 2E		5	7,24	4	25				
R6	2B to 2E		4	7,24	5	25				
R7	2C to 2E		24	4,5	7	25				
R8	2D to 2E		7	4,5	24	25				
R9	3A to 3E		9	30,31	8	29				
R10	3B to 3E		8	30,31	9	29				
R11	3C to 3E		31	8,9	30	29				
R12	3D to 3E		30	8,9	31	29				
R13	4A to 4E		27	6,26	28	38				
R14	4B to 4E		28	6,26	27	38				
R15	4C to 4E		26	28,27	6	38				
R16	4D to 4E		6	28,27	26	38				
R17	5A to 5E	Propagation delay time (t_{PLH}), C_L = 50 pF	35	15,12	16	36	3.5	15.0	ns	18,20

* TABLE XVI. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
R18	5B to 5E	Propagation delay time (t_{PLH}), C_L = 50 pF	16	15,12	35	36	3.5	15.0	ns	18,20
R19	5C to 5E		12	16,35	15	36				
R20	5D to 5E		15	16,35	12	36				
R21	6A to 6E		34	14,13	32	33				
R22	6B to 6E		32	14,13	34	33				
R23	6C to 6E		13	32,34	14	33				
R24	6D to 6E		14	32,34	13	33	3.5	15.0		18,20
* R25	1A to 1F		23	2,22	3	17	5.5	21.0		18,19
R26	1B to 1F		3	2,22	23	17				
R27	1C to 1F		22	3,23	2	17				
R28	1D to 1F		2	3,23	22	17				
R29	2A to 2F		5	7,24	4	19				
R30	2B to 2F		4	7,24	5	19				
R31	2C to 2F		24	4,5	7	19				
R32	2D to 2F		7	4,5	24	19				
R33	3A to 3F		9	30,31	8	39				
* R34	3B to 3F	Propagation delay time (t_{PLH}), C_L = 50 pF	8	30,31	9	39	5.5	21.0	ns	18,19

Source: https://assist.dla.mil -- Downloaded: 202640-20T15:49Z
Check the source to verify that this is the current version before use.

* TABLE XVI. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

	Test condition	Parameter		Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Initial limits 25° ±5°C Ambient Min	Initial limits 25° ±5°C Ambient Max	Unit	Test figure
*	R35	3C to 3F	Propagation delay time (t_{PLH}), C_L = 50 pF	30	8,9	31	39	5.5	21.0	ns	18,19
	R36	3D to 3F		31	8,9	30	39				
	R37	4A to 4F		27	6,26	28	18				
	R38	4B to 4F		28	6,26	27	18				
	R39	4C to 4F		26	28,27	6	18				
	R40	4D to 4F		6	28,27	26	18				
	R41	5A to 5F		35	15,12	16	40				
	R42	5B to 5F		16	15,12	35	40				
	R43	5C to 5F		12	16,35	15	40				
	R44	5D to 5F		15	16,35	12	40				
	R45	6A to 6F		34	14,13	32	37				
	R46	6B to 6F		32	14,13	34	37				
	R47	6C to 6F		13	32,34	14	37				
*	R48	6D to 6F	Propagation delay time (t_{PLH}), C_L = 50 pF	14	32,34	13	37	5.5	21.0		18,19
	R49	1A to 1E	Propagation delay time (t_{PLH}), C_L = 150 pF	23	2,22	3	21	5.0	18.0		18,20
	R50	1B to 1E	Propagation delay time (t_{PLH}), C_L = 150 pF	3	2,22	23	21	5.0	18.0	ns	18,20

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:49Z
Check the source to verify that this is the current version before use.

* TABLE XVI. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
R51	1C to 1E	Propagation delay time (t_{PLH}), C_L = 150 pF	22	3,23	2	21	5.0	18.0	ns	18,20
R52	1D to 1E	↓	2	3,23	22	21	↓	↓	↓	↓
R53	2A to 2E		5	7,24	4	25				
R54	2B to 2E		4	7,24	5	25				
R55	2C to 2E		24	4,5	7	25				
R56	2D to 2E		7	4,5	24	25				
R57	3A to 3E		9	30,31	8	29				
R58	3B to 3E		8	30,31	9	29				
R59	3C to 3E		30	8,9	31	29				
R60	3D to 3E		31	8,9	30	29				
R61	4A to 4E		27	6,26	28	38				
R62	4B to 4E		28	6,26	27	38				
R63	4C to 4E		26	28,27	6	38				
R64	4D to 4E		6	28,27	26	38				
R65	5A to 5E		35	15,12	16	36				
R66	5B to 5E		16	15,12	35	36				
R67	5C to 5E	Propagation delay time (t_{PLH}), C_L = 150 pF	12	16,35	15	36	5.0	18.0	ns	18,20

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:49Z
Check the source to verify that this is the current version before use.

* TABLE XVI. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
R68	5D to 5E	Propagation delay time (t_{PLH}), C_L = 150 pF	15	16,35	12	36	5.0	18.0	ns	18,20
R69	6A to 6E		34	14,13	32	33				
R70	6B to 6E		32	14,13	34	33				
R71	6C to 6E		13	32,34	14	33				
R72	6D to 6E		14	32,34	13	33	5.0	18.0		18,20
R73	1A to 1F		23	2,22	3	17	7.0	27.0		18,19
R74	1B to 1F		3	2,22	23	17				
R75	1C to 1F		22	3,23	2	17				
R76	1D to 1F		2	3,23	22	17				
R77	2A to 2F		5	7,24	4	19				
R78	2B to 2F		4	7,24	5	19				
R79	2C to 2F		24	4,5	7	19				
R80	2D to 2F		7	4,5	24	19				
R81	3A to 3F		9	30,31	8	39				
R82	3B to 3F		8	30,31	9	39				
R83	3C to 3F		30	8,9	31	39				
R84	3D to 3F	Propagation delay time (t_{PLH}), C_L = 150 pF	31	8,9	30	39	7.0	27.0	ns	18,19

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:49Z
Check the source to verify that this is the current version before use.

* TABLE XVI. Propagation delay time test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
R85	4A to 4F	Propagation delay time (t_{PLH}), C_L = 150 pF	27	6,26	28	18	7.0	27.0	ns	18,19
R86	4B to 4F	↓	28	6,26	27	18	↓	↓	↓	↓
R87	4C to 4F	↓	26	28,27	6	18	↓	↓	↓	↓
R88	4D to 4F	↓	6	28,27	26	18	↓	↓	↓	↓
R89	5A to 5F	↓	35	15,12	16	40	↓	↓	↓	↓
R90	5B to 5F	↓	16	15,12	35	40	↓	↓	↓	↓
R91	5C to 5F	↓	12	16,35	15	40	↓	↓	↓	↓
R92	5D to 5F	↓	15	16,35	12	40	↓	↓	↓	↓
R93	6A to 6F	↓	34	14,13	32	37	↓	↓	↓	↓
R94	6B to 6F	↓	32	14,13	34	37	↓	↓	↓	↓
R95	6C to 6F	↓	13	32,34	14	37	↓	↓	↓	↓
R96	6D to 6F	Propagation delay time (t_{PLH}), C_L = 150 pF	14	32,34	13	37	7.0	27.0	↓	18,19
S1	1A to 1E	Propagation delay time (t_{PHL}), C_L = 50 pF	23	2,22	3	21	3.5	10.5	↓	18,20
S2	1B to 1E	↓	3	2,22	23	21	↓	↓	↓	↓
S3	1C to 1E	↓	22	3,23	2	21	↓	↓	↓	↓
S4	1D to 1E	Propagation delay time (t_{PHL}), C_L = 50 pF	2	3,23	22	21	3.5	10.5	ns	18,20

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:49Z
Check the source to verify that this is the current version before use.

TABLE XVI. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
S5	2A to 2E	Propagation delay time (t_{PHL}), C_L = 50 pF	5	7,24	4	25	3.5	10.5	ns	18,20
S6	2B to 2E		4	7,24	5	25				
S7	2C to 2E		24	4,5	7	25				
S8	2D to 2E		7	4,5	24	25				
S9	3A to 3E		9	30,31	8	29				
S10	3B to 3E		8	30,31	9	29				
S11	3C to 3E		31	8,9	30	29				
S12	3D to 3E		30	8,9	31	29				
S13	4A to 4E		27	6,26	28	38				
S14	4B to 4E		28	6,26	27	38				
S15	4C to 4E		26	28,27	6	38				
S16	4D to 4E		6	28,27	26	38				
S17	5A to 5E		35	15,12	16	36				
S18	5B to 5E		16	15,12	35	36				
S19	5C to 5E		12	16,35	15	36				
S20	5D to 5E		15	16,35	12	36				
S21	6A to 6E	Propagation delay time (t_{PHL}), C_L = 50 pF	34	14,13	32	33	3.5	10.5	ns	18,20

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:49Z
Check the source to verify that this is the current version before use.

* TABLE XVI. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
S22	6B to 6E	Propagation delay time (t_{PHL}), C_L = 50 pF	32	14,13	34	33	3.5	10.5	ns	18,20
S23	6C to 6E		13	32,34	14	33	3.5	10.5		18,20
S24	6D to 6E		14	32,34	13	33	3.5	10.5		18,20
S25	1A to 1F		23	2,22	3	17	5.5	17.0		18,19
S26	1B to 1F		3	2,22	23	17				
S27	1C to 1F		22	3,23	2	17				
S28	1D to 1F		2	3,23	22	17				
S29	2A to 2F		5	7,24	4	19				
S30	2B to 2F		4	7,24	5	19				
S31	2C to 2F		24	4,5	7	19				
S32	2D to 2F		7	4,5	24	19				
S33	3A to 3F		9	30,31	8	39				
S34	3B to 3F		8	30,31	9	39				
S35	3C to 3F		30	8,9	31	39				
S36	3D to 3F		31	8,9	30	39				
S37	4A to 4F		27	6,26	28	18				
S38	4B to 4F	Propagation delay time (t_{PHL}), C_L = 50 pF	28	6,26	27	18	5.5	17.0	ns	18,19

Source: https://assist.dla.mil -- Downloaded: 20260-20T15:49Z
Check the source to verify that this is the current version before use.

TABLE XVI. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
S39	4C to 4F	Propagation delay time (t_{PHL}), C_L = 50 pF	26	28,27	6	18	5.5	17.0	ns	18,19
S40	4D to 4F	↓	6	28,27	26	18	↓	↓	↓	↓
S41	5A to 5F	↓	35	15,12	16	40	↓	↓	↓	↓
S42	5B to 5F	↓	16	15,12	35	40	↓	↓	↓	↓
S43	5C to 5F	↓	12	16,35	15	40	↓	↓	↓	↓
S44	5D to 5F	↓	15	16,35	12	40	↓	↓	↓	↓
S45	6A to 6F	↓	34	14,13	32	37	↓	↓	↓	↓
S46	6B to 6F	↓	32	14,13	34	37	↓	↓	↓	↓
S47	6C to 6F	↓	13	32,34	14	37	↓	↓	↓	↓
S48	6D to 6F	Propagation delay time (t_{PHL}), C_L = 50 pF	14	32,34	13	37	5.5	17.0	↓	18,19
S49	1A to 1E	Propagation delay time (t_{PHL}), C_L = 150 pF	23	2,22	3	21	4.5	18.0	↓	18,20
S50	1B to 1E	↓	3	2,22	23	21	↓	↓	↓	↓
S51	1C to 1E	↓	22	3,23	2	21	↓	↓	↓	↓
S52	1D to 1E	↓	2	3,23	22	21	↓	↓	↓	↓
S53	2A to 2E	↓	5	7,24	4	25	↓	↓	↓	↓
S54	2B to 2E	Propagation delay time (t_{PHL}), C_L = 150 pF	4	7,24	5	25	4.5	18.0	ns	18,20

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:49Z
Check the source to verify that this is the current version before use.

* TABLE XVI. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
S55	2C to 2E	Propagation delay time (t_{PHL}), C_L = 150 pF	24	4,5	7	25	4.5	18.0	ns	18,20
S56	2D to 2E	↓	7	4,5	24	25	↓	↓	↓	↓
S57	3A to 3E		9	30,31	8	29				
S58	3B to 3E		8	30,31	9	29				
S59	3C to 3E		30	8,9	31	29				
S60	3D to 3E		31	8,9	30	29				
S61	4A to 4E		27	6,26	28	38				
S62	4B to 4E		28	6,26	27	38				
S63	4C to 4E		26	28,27	6	38				
S64	4D to 4E		6	28,27	26	38				
S65	5A to 5E		35	15,12	16	36				
S66	5B to 5E		16	15,12	35	36				
S67	5C to 5E		12	16,35	15	36				
S68	5D to 5E		15	16,35	12	36				
S69	6A to 6E		34	14,13	32	33				
S70	6B to 6E		32	14,13	34	33				
S71	6C to 6E	Propagation delay time (t_{PHL}), C_L = 150 pF	13	32,34	14	33	4.5	18.0	ns	18,20

Source: https://assist.dla.mil -- Downloaded: 2026/10-20T15:49Z
Check the source to verify that this is the current version before use.

TABLE XVI. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

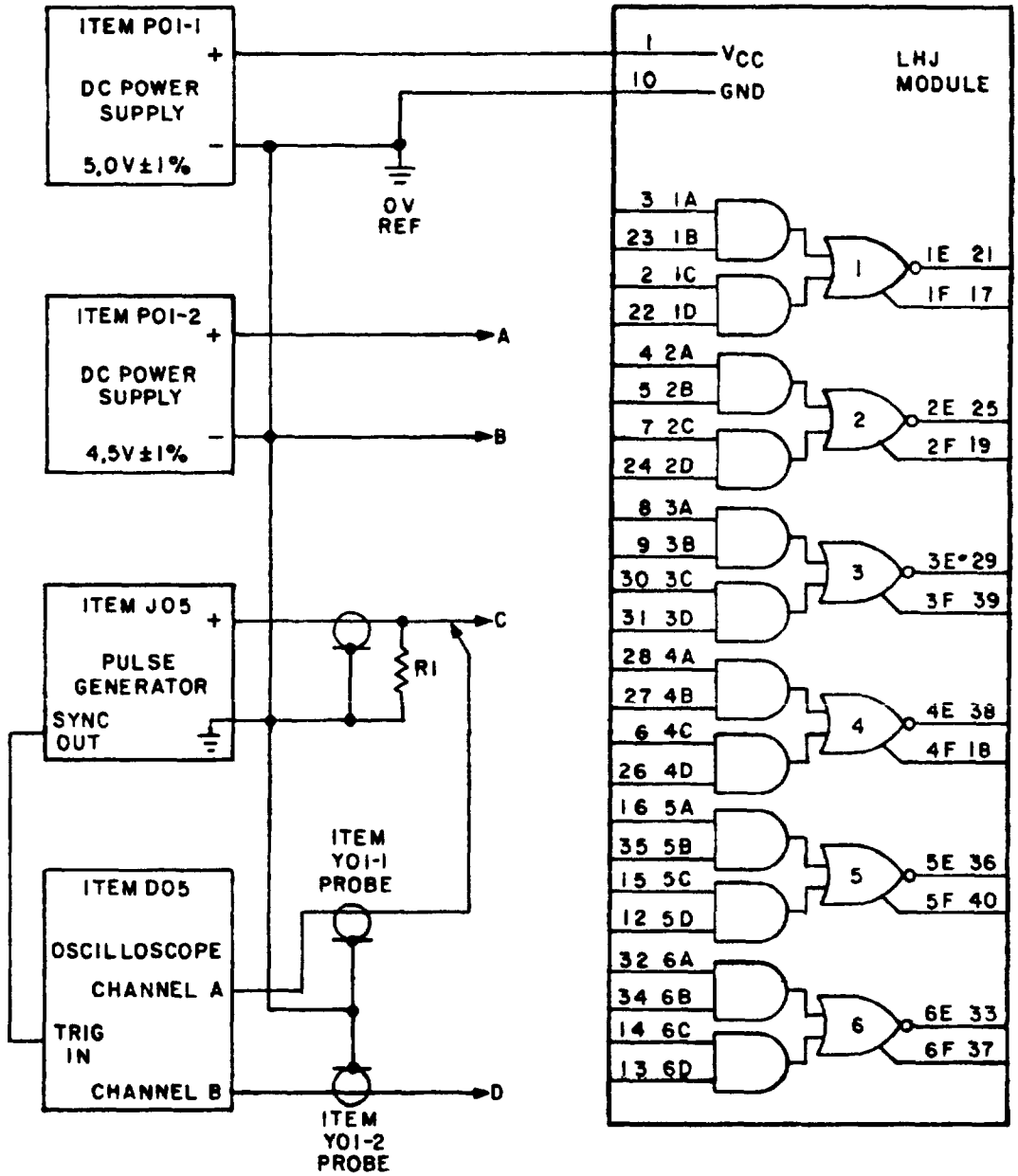
Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
S72	6D to 6E	Propagation delay time (t_{PHL}), C_L = 150 pF	14	32,34	13	33	4.5	18.0	ns	18,20
S73	1A to 1F	Propagation delay time (t_{PHL}), C_L = 150 pF	23	2,22	3	17	6.5	24 5		18,19
S74	1B to 1F		3	2,22	23	17				
S75	1C to 1F		22	3,23	2	17				
S76	1D to 1F		2	3,23	22	17				
S77	2A to 2F		5	7,24	4	19				
S78	2B to 2F		4	7,24	5	19				
S79	2C to 2F		24	4,5	7	19				
S80	2D to 2F		7	4,5	24	19				
S81	3A to 3F		9	30,31	8	39				
S82	3B to 3F		8	30,31	9	39				
S83	3C to 3F		30	8,9	31	39				
S84	3D to 3F		31	8,9	30	39				
S85	4A to 4F		27	6,26	28	18				
S86	4B to 4F		28	6,26	27	18				
S87	4C to 4F	Propagation delay time (t_{PHL}), C_L = 150 pF	26	28,27	6	18	6.5	24.5	ns	18,19

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:49Z
Check the source to verify that this is the current version before use.

TABLE XVI. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
S88	4D to 4F	Propagation delay time (t_{PHL}), C_L = 150 pF	6	28,27	26	18	6.5	24.5	ns	18,19
S89	5A to 5F	↓	35	15,12	16	40	↓	↓	↓	↓
S90	5B to 5F		16	15,12	35	40				
S91	5C to 5F		12	16,35	15	40				
S92	5D to 5F		15	16,35	12	40				
S93	6A to 6F		34	14,13	32	37				
S94	6B to 6F		32	14,13	34	37				
S95	6C to 6F		13	32,34	14	37				
S96	6D to 6F	Propagation delay time (t_{PHL}), C_L = 150 pF	14	32,34	13	37	6.5	24.5	ns	18,19

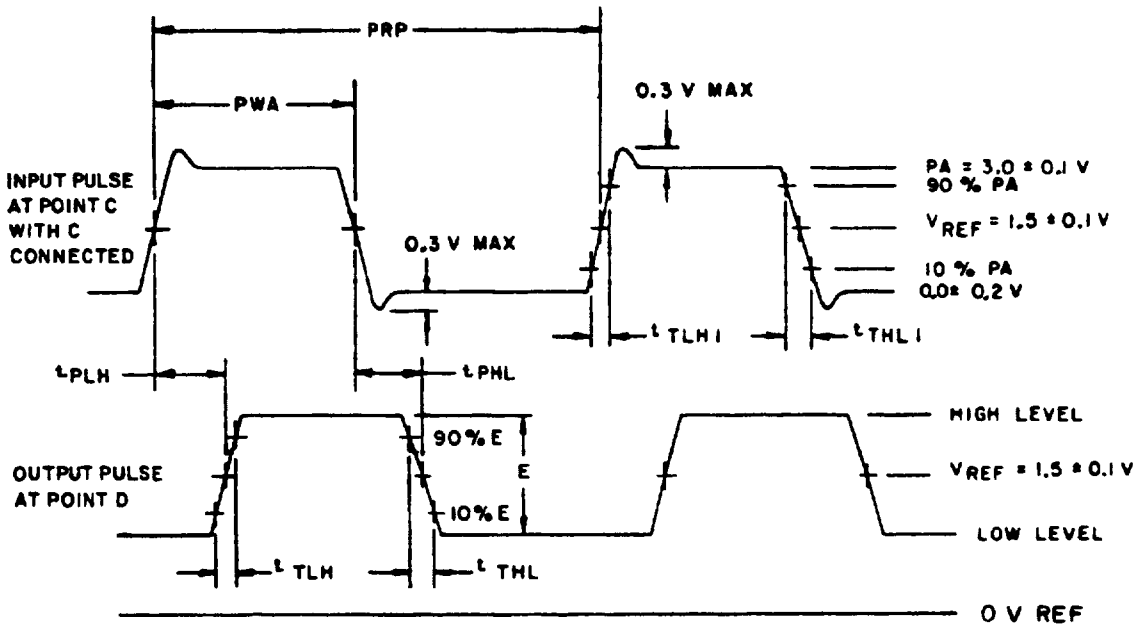
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:49Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = 51Ω ±5%, 1/2 W.
2. Connect logic load per figure 12 to both E and F outputs.

* FIGURE 18. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for tables XV and XVI.

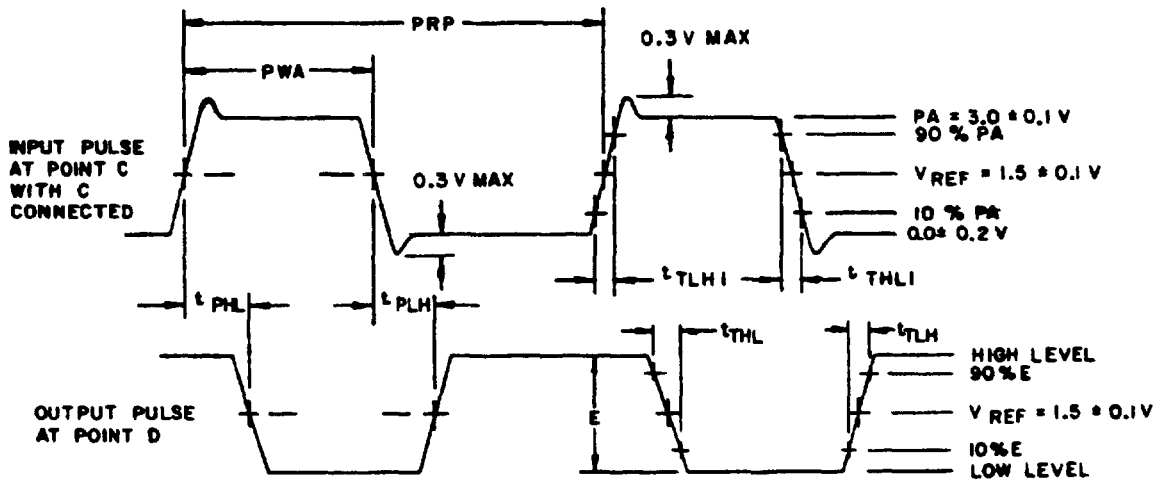
Source: https://assist.dla.mil -- Downloaded: 202610-20T15:49Z
Check the source to verify that this is the current version before use.



NOTES:
1. Input pulse parameters:
   $t_{TLH1} = t_{THL1} = 4 \pm 1$ ns
   PWA = 500 ±50 ns
   PRP = 1,000 ±100 ns.
2. Before making the measurements specified in tables XV and XVI verify that the output pulse high level is $\geq$ 2.5 V and the low level is $\leq$ 0.5 V.
3. The lead length of electrical conductors between resistor R1 and input pins; between probes YO1-1 and YO1-2 and output pins, and between load circuits and output pins, shall be less than 1 inch (25.4 mm).
4. All voltages are referenced to ground.

* FIGURE 19 Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XV and XVI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:49Z
Check the source to verify that this is the current version before use.



NOTES:
1. Input pulse parameters:
   $t_{TLH1} = t_{THL1} = 4 \pm 1$ ns
   * PWA = 500 ±50 ns
   PRP = 1,000 ±50 ns.
2. Before making the measurements specified in table XVI verify that the output pulse high level is $\geq$ 2.5 V and the low level is $\leq$ 0.5 V
3. The lead length of electrical conductors between resistor R1 and input pin, between probes Y01-1 and Y01-2 and input and output pins, and between load circuit and output pin, shall be less than 1 inch (25.4 mm).
4. All voltages are referenced to ground.

* FIGURE 20. Propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for table XVI.

Source: https://assist.dla.mil -- Downloaded: 202640-20T15:49Z
Check the source to verify that this is the current version before use.

TABLE XVII. Output crosstalk (low-level and high-level) test.

	Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Initial limits 25° ±5°C Ambient Min	Initial limits 25° ±5°C Ambient Max	Unit	Test figure
*	T1	1E Output crosstalk, low level	3,23,5, 9,27, 35,34	2,22,7, 24,30, 31,6, 26,15, 12,14, 13	4,8,28, 16,32	21	0	750	mV	21,22
	T2	2E	4,5,23, 9,27, 35,34		3,8,28, 16,32	25				
	T3	3E	8,9,23, 5,27, 35,34		3,4,28, 16,32	29				
	T4	4E	28,27, 23,5,9, 35,34		3,4,8, 16,32	38				
	T5	5E	16,35, 23,5,9, 27,34		3,4,8, 28,32	36				
*	T6	6E	32,34, 23,5,9, 27,35	2,22,7, 24,30, 31,6, 26,15, 12,14, 13	3,4,8, 28,16	33		750		
*	T7	1F Output crosstalk, low level	5,9,27, 35,34	3,23,2, 22,7, 24,30, 31,6, 26,15, 12,14, 13	4,8,28, 16,32	17	0	1000	mV	21,22

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:49Z
Check the source to verify that this is the current version before use.

TABLE XVII. Output crosstalk (low-level and high-level) test - Continued.

	Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
*	T8	2F Output crosstalk, low level	23,9, 27,35, 34	4,5,7, 24,2, 22,30, 31,6, 26,15, 12,14, 13	3,8,28, 16,32	19	0	1000	mV	21,22
	T9	3F	23,5, 27,35, 34	8,9,2, 22,7, 24,30, 31,6, 26,15, 12,14, 13	3,4,28, 16,32	39				
	T10	4F	23,5,9, 35,34	28,27, 2,22,7, 24,30, 31,6, 26,15, 12,14, 13	3,4,8, 16,32	18				
*	T11	5F Output crosstalk, low level	23,5,9, 27,34	16,35, 2,22,7, 24,30, 31,6, 26,15, 12,14, 13	3,4,8, 28,32	40	0	1000	mV	21,22

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:49Z
Check the source to verify that this is the current version before use.

TABLE XVII. Output crosstalk (low-level and high-level) test - Continued.

	Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
*	T12	6F Output crosstalk, low level	23,5,9, 27,35	32,34, 2,22,7, 24,30, 31,6, 26,15, 12,14, 13	3,4,8, 28,16	37	0	1000	mV	21,22
*	T13	1F	5,35, 34	3,23,2, 22,7, 24,30, 31,8,9, 28,27, 6,26, 15,12, 14,13	4,16, 32	17		500		
*	T14	3F	5,35, 34	3,23,2, 22,7, 24,30, 31,8,9, 28,27, 6,26, 15,12, 14,13	4,16, 32	39		500		
*	T15	4F Output crosstalk, low level	5,35, 34	3,23,2, 22,7, 24,30, 31,8,9, 28,27, 6,26, 15,12, 14,13	4,16, 32	18	0	500	mV	21,22

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:49Z
Check the source to verify that this is the current version before use.

TABLE XVII. Output crosstalk (low-level and high-level) test - Continued.

	Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
*	T16	2F Output crosstalk, low level	23,9, 27	2,22,4, 5,7,24, 30,31, 16,35, 15,12, 32,34, 14,13	3,8,28	19	0	500	mV	21,22
*	T17	5F Output crosstalk, low level	23,9, 27	2,22,4, 5,7,24, 30,31, 16,35, 15,12, 32,34, 14,13	3,8,28	40		500		
*	T18	6F Output crosstalk, low level	23,9, 27	2,22,4, 5,7,24, 30,31, 16,35, 15,12, 32,34, 14,13	3,8,28	37		500		
	U1	1E Output crosstalk, high level	5,9,27, 35,34	3,23,2, 22,7, 24,30, 31,6, 26,15, 12,14, 13	4,8,28, 16,32	21	0	400	mV	21,22

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:49Z
Check the source to verify that this is the current version before use.

TABLE XVII. Output crosstalk (low-level and high-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
U2	2E Output crosstalk, high level	23,9, 27,35, 34	4,5,7, 24,2, 22,30, 31,6, 16,15, 12,14, 13	3,8,28, 16,32	25	0	400	mV	21,22
U3	3E	23,5, 27,35, 34	8,9,2, 22,7, 24,30, 31,6, 26,15, 12,14, 13	3,4,28, 16,32	29				
U4	4E	23,5,9, 35,34	28,27, 2,22,7, 24,30, 31,6, 26,15, 12,14, 13	3,4,8, 16,32	38				
U5	5E Output crosstalk, high level	23,5,9, 27,34	16,35, 2,22,7, 24,30, 31,6, 26,15, 12,14, 13	3,4,8, 28,32	36	0	400	mV	21,22

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:49Z
Check the source to verify that this is the current version before use.

TABLE XVII. Output crosstalk (low-level and high-level) test - Continued.

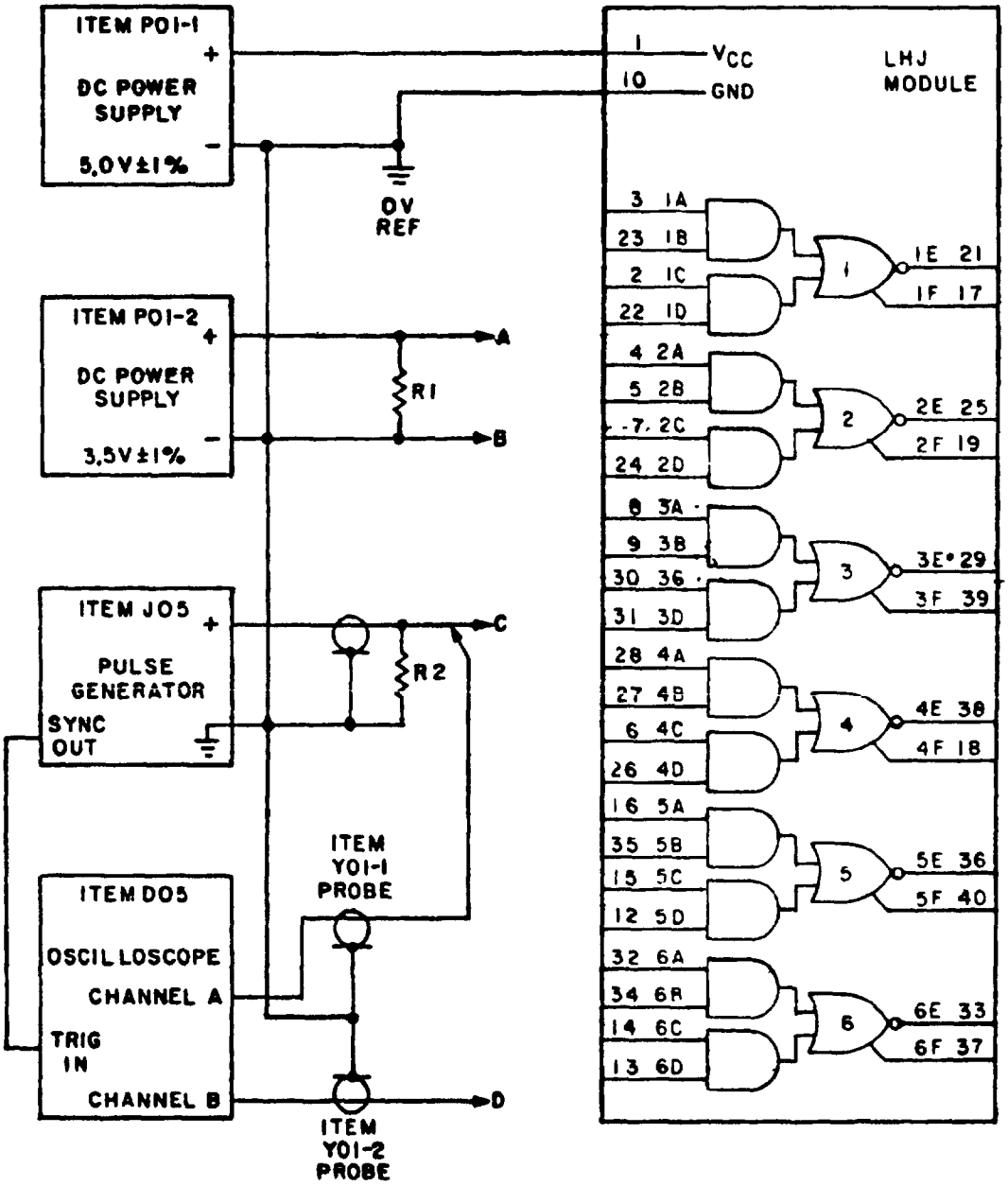
Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
U6	6E	Output crosstalk, high level	23,5,9, 27,35	32,34, 2,22,7, 24,30, 31,6, 26,15, 12,14, 13	3,4,8, 28,16	33	0	400	mV	21,22
U7	1F		3,23,5, 9,27, 35,34	2,22,7, 24,30, 31,6, 26,15, 12,14, 13	4,8,28, 16,32	17				
U8	2F		4,5,23, 9,27, 35,34	2,22,7, 24,30, 31,6, 26,15, 12,14, 13	3,8,28, 16,32	19				
U9	3F		8,9,23, 5,27, 35,34	2,22,7, 24,30, 31,6, 26,15, 12,14, 13	3,4,28, 16,32	39				
U10	4F	Output crosstalk, high level	28,27, 23,5,9, 35,34	2,22,7, 24,30, 31,6, 26,15, 12,14, 13	3,4,8, 16,32	18	0	400	mV	21,22

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:49Z
Check the source to verify that this is the current version before use.

TABLE XVII. Output crosstalk (low-level and high-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
U11	5F Output crosstalk, high level	16,35, 23,5,9, 27,34	2,22,7, 24,30, 31,6, 26,15, 12,14, 13	3,4,8, 28,32	40	0	400	mV	21,22
U12	6F Output crosstalk, high level	32,34, 23,5,9, 27,35	2,22,7, 24,30, 31,6, 26,15, 12,14, 13	3,4,8, 28,16	37	0	400	mV	21,22

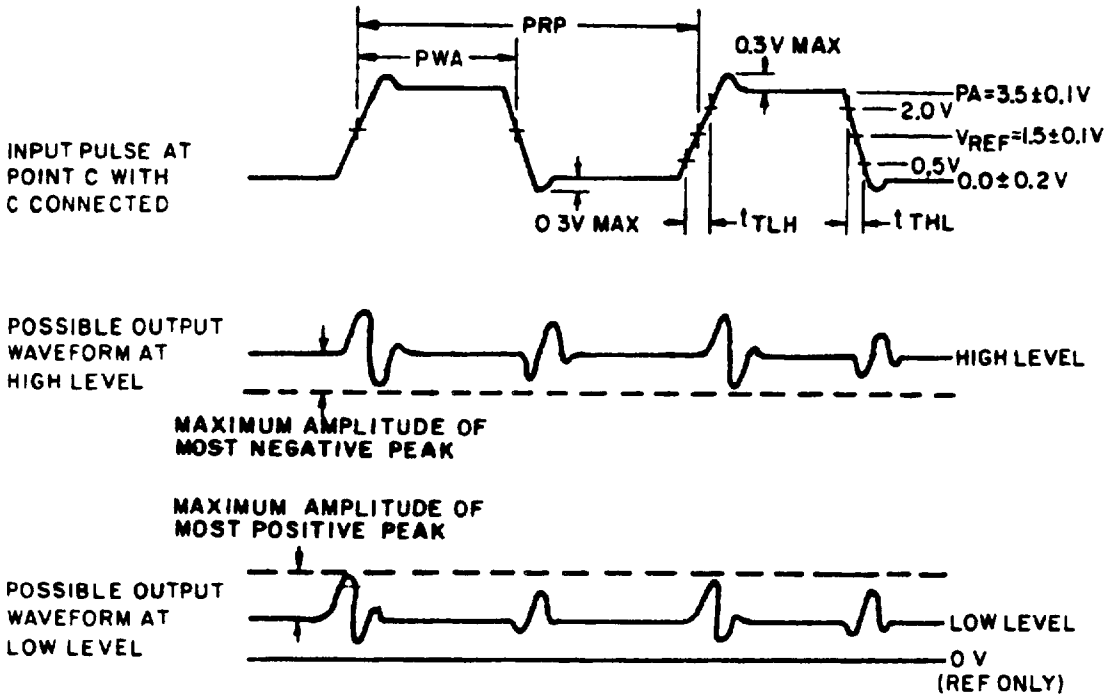
Source: https://assist.dla.mil -- Downloaded: 20260-20T15:49Z
Check the source to verify that this is the current version before use.



NOTES
1 R1 = R2 = 51Ω ±5%, 1/2 W
2 Load all outputs with logic load per figure 12 (C_L = 50 pF).

FIGURE 21 Output crosstalk (low-level and high-level) test circuit for table XVII

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:49Z
Check the source to verify that this is the current version before use.



NOTES:
1. Input pulse parameters.
   $t_{THL} = t_{TLH}$ = 3.5 ±1.5 ns (verified at the module pin)
   PWA = 500 ±50 ns
   PRP = 1,000 ±100 ns.
2. The lead length of electrical conductors between resistor R2 and input pins, between probes YO1-1 and YO1-2 and input and output pins; between load circuits and output pins; and between ground leads of probes YO1-1 and YO1-2 to test fixture and module ground pins, shall be less than 1 inch (25.4 mm).

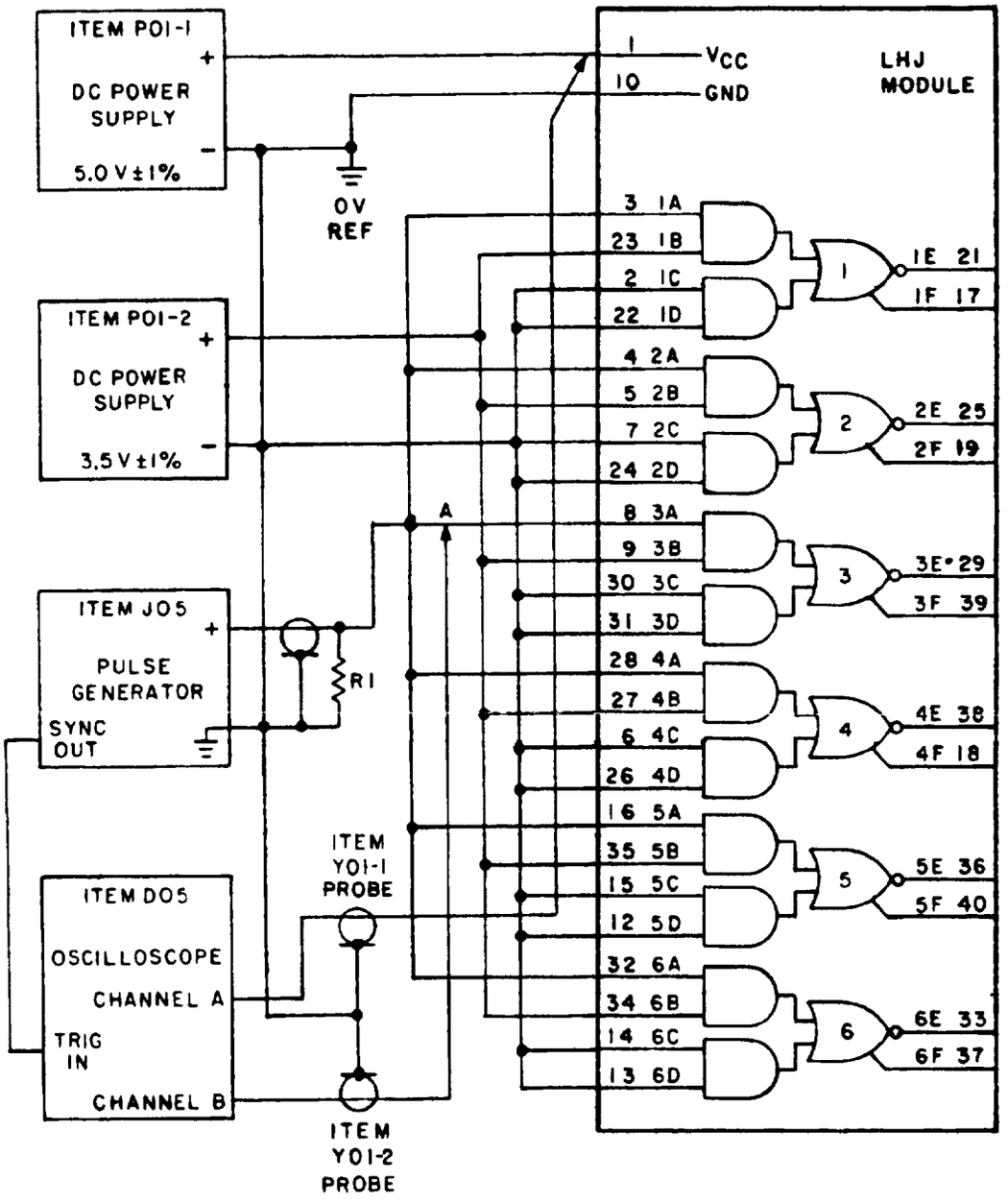
FIGURE 22 Output crosstalk (low-level and high-level) test circuit waveforms for table XVII.

Source: https://assist.dla.mil -- Downloaded: 202640-20T15:49Z
Check the source to verify that this is the current version before use.

TABLE XVIII. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25° ±5°C Ambient		Unit	Test figure
		Min	Max		
V	Power supply transient amplitude	0	500	mV	23,24

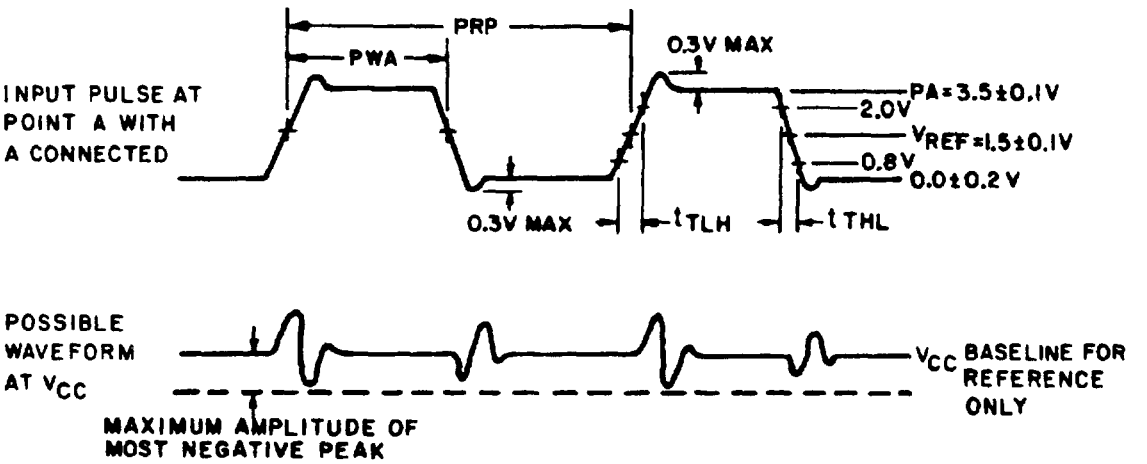
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:49Z
Check the source to verify that this is the current version before use.



NOTE: R1 = 51Ω ±5%, 1/2 W.

FIGURE 23. Power supply transient amplitude test circuit for table XVIII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:49Z
Check the source to verify that this is the current version before use.



NOTE: Input pulse parameters:
$t_{THL} = t_{TLH} = 3.5 \pm 1.5$ ns (verified at the module pin)
PWA = 500 ±50 ns
PRP = 1,000 ±100 ns.

FIGURE 24. Power supply transient amplitude test circuit waveforms for table XVIII.

Source: https://assist.dla.mil -- Downloaded: 202640-20T15:49Z
Check the source to verify that this is the current version before use.

TABLE XIX. Power supply filter capacitance test.

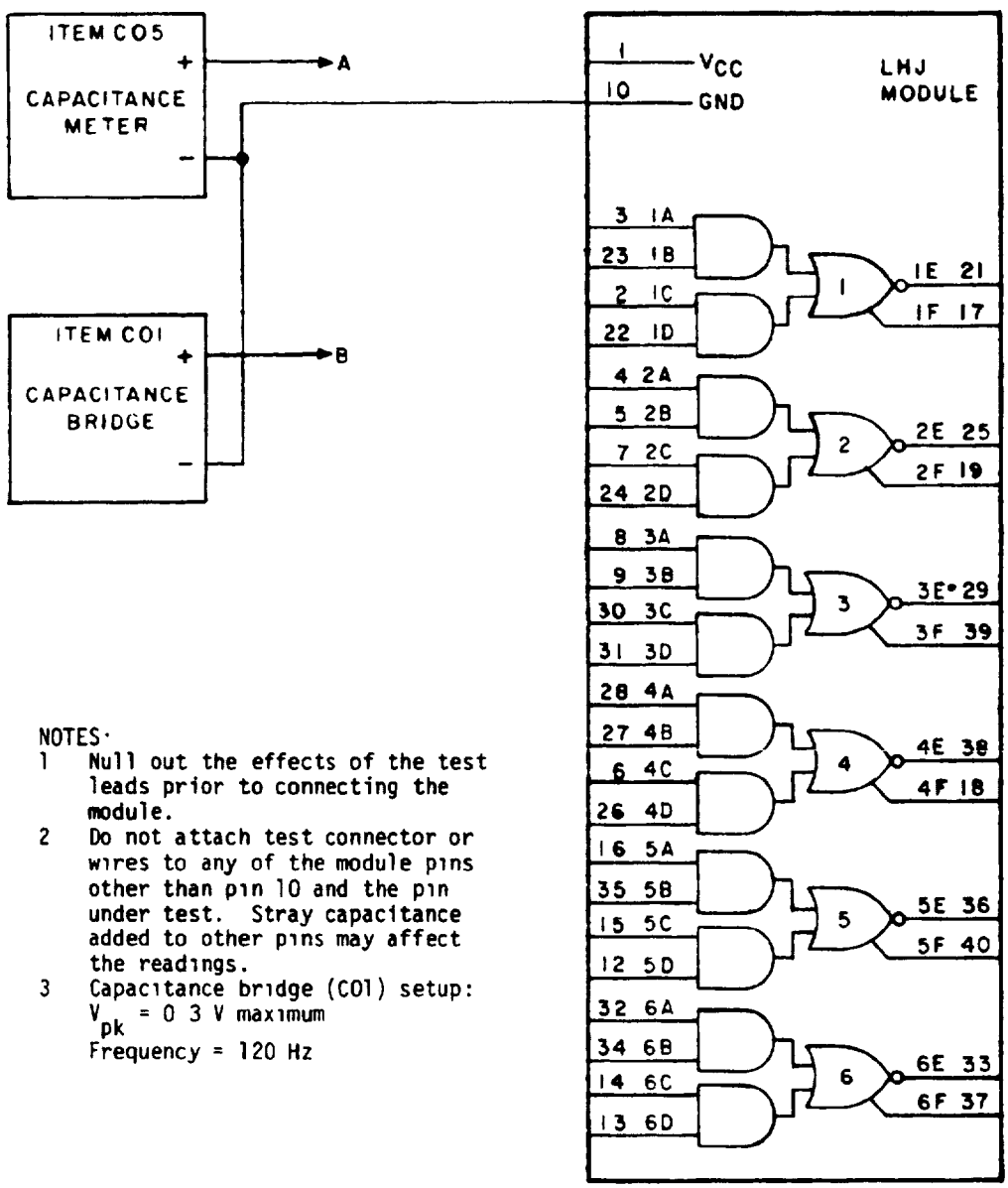
Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
W	Power supply filter capacitance	Open	1	1.6	3.0	µF	25

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:49Z
Check the source to verify that this is the current version before use.

TABLE XX. Input capacitance test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
X1	1A Input capacitance	3	Open	2.0	15.0	pF	25
X2	1B	23					
X3	1C	2					
X4	1D	22					
X5	2A	4					
X6	2B	5					
X7	2C	7					
X8	2D	24					
X9	3A	8					
X10	3B	9					
X11	3C	30					
X12	3D	31					
X13	4A	28					
X14	4B	27					
X15	4C	6					
X16	4D	26					
X17	5A	16					
X18	5B	35					
X19	5C	15					
X20	5D	12					
X21	6A	32					
X22	6B	34					
X23	6C	14					
X24	6D Input capacitance	13	Open	2.0	15.0	pF	25

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T15:49Z
Check the source to verify that this is the current version before use.



NOTES:
1. Null out the effects of the test leads prior to connecting the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
3. Capacitance bridge (CO1) setup:
   V_{pk} = 0.3 V maximum
   Frequency = 120 Hz

FIGURE 25. Power supply filter capacitance and input capacitance test circuit for tables XIX and XX.

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:49Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY
NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D.C. 20360

POSTAGE AND FEES PAID
NAVY DEPARTMENT
DOD 316

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

COMMANDER
NAVAL ELECTRONIC SYSTEMS COMMAND
DEFENSE STANDARDIZATION PROGRAM BRANCH
DEPARTMENT OF THE NAVY
WASHINGTON, D.C. 20360

FOLD

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:49Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No 22-R255

INSTRUCTIONS The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government Fold on lines on reverse side, staple in corner, and send to preparing activity Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity

DOCUMENT IDENTIFIER AND TITLE

MIL-M-28787/64A (EC)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3 IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4 REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

S/N 0102-014-1802

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:49Z
Check the source to verify that this is the current version before use.